tropical: FractionalKnapsack

2026-10-25T15:33:20

1 Key results

This report covers 240 runs (5 seeds; 0 diverged) under frozen hash `3b4ea5b21c9e`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6015	[0.597, 0.606]	0.00584	5	0
adaptive	pointer	0.01	0.6021	[0.599, 0.607]	0.00551	5	0
adaptive	pointer	0.1	0.6012	[0.597, 0.606]	0.00581	5	0
adaptive	pointer	1	0.6021	[0.598, 0.607]	0.00606	5	0
adaptive	pointer	10	0.6022	[0.599, 0.607]	0.00519	5	0
adaptive	pointer	100	0.6012	[0.597, 0.606]	0.00555	5	0
adaptive	pointer	1000	0.5968	[0.593, 0.601]	0.00528	5	0
adaptive	pointer	1e+04	0.5975	[0.592, 0.603]	0.00713	5	0
adaptive	seq2seq	0	0.5368	[0.529, 0.544]	0.0101	5	0
adaptive	seq2seq	0.01	0.5402	[0.531, 0.549]	0.0125	5	0
adaptive	seq2seq	0.1	0.5404	[0.532, 0.551]	0.0123	5	0
adaptive	seq2seq	1	0.5398	[0.531, 0.55]	0.0123	5	0
adaptive	seq2seq	10	0.5392	[0.531, 0.55]	0.012	5	0
adaptive	seq2seq	100	0.5375	[0.531, 0.547]	0.0103	5	0
adaptive	seq2seq	1000	0.5451	[0.531, 0.559]	0.0191	5	0
adaptive	seq2seq	1e+04	0.5454	[0.542, 0.549]	0.00402	5	0
tropical	pointer	0	0.6633	[0.631, 0.694]	0.0376	5	0
tropical	pointer	0.01	0.6578	[0.632, 0.675]	0.0298	5	0
tropical	pointer	0.1	0.6585	[0.631, 0.679]	0.0328	5	0
tropical	pointer	1	0.6638	[0.637, 0.683]	0.0313	5	0
tropical	pointer	10	0.6615	[0.631, 0.685]	0.0368	5	0
tropical	pointer	100	0.6595	[0.634, 0.679]	0.0287	5	0
tropical	pointer	1000	0.6571	[0.628, 0.676]	0.0326	5	0
tropical	pointer	1e+04	0.6562	[0.631, 0.672]	0.0277	5	0
tropical	seq2seq	0	0.7213	[0.695, 0.747]	0.033	5	0
tropical	seq2seq	0.01	0.7173	[0.688, 0.746]	0.0383	5	0
tropical	seq2seq	0.1	0.7202	[0.694, 0.748]	0.0365	5	0
tropical	seq2seq	1	0.7165	[0.689, 0.744]	0.0367	5	0
tropical	seq2seq	10	0.718	[0.694, 0.745]	0.0327	5	0
tropical	seq2seq	100	0.718	[0.692, 0.746]	0.0352	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1000	0.6992	[0.677, 0.724]	0.0312	5	0
tropical	seq2seq	1e+04	0.6402	[0.608, 0.676]	0.0435	5	0
vanilla	pointer	0	0.6045	[0.601, 0.608]	0.00456	5	0
vanilla	pointer	0.01	0.6076	[0.605, 0.61]	0.00354	5	0
vanilla	pointer	0.1	0.6075	[0.605, 0.61]	0.00351	5	0
vanilla	pointer	1	0.6075	[0.605, 0.61]	0.00379	5	0
vanilla	pointer	10	0.6075	[0.604, 0.611]	0.00399	5	0
vanilla	pointer	100	0.6069	[0.604, 0.609]	0.00348	5	0
vanilla	pointer	1000	0.6062	[0.603, 0.609]	0.00414	5	0
vanilla	pointer	1e+04	0.6044	[0.6, 0.608]	0.00476	5	0
vanilla	seq2seq	0	0.5548	[0.528, 0.581]	0.0349	5	0
vanilla	seq2seq	0.01	0.5662	[0.542, 0.582]	0.0273	5	0
vanilla	seq2seq	0.1	0.5669	[0.542, 0.583]	0.0274	5	0
vanilla	seq2seq	1	0.5669	[0.542, 0.583]	0.0274	5	0
vanilla	seq2seq	10	0.5724	[0.561, 0.58]	0.0123	5	0
vanilla	seq2seq	100	0.5612	[0.543, 0.572]	0.0199	5	0
vanilla	seq2seq	1000	0.5524	[0.536, 0.563]	0.0183	5	0
vanilla	seq2seq	1e+04	0.5182	[0.505, 0.534]	0.0173	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	5.591×10^{-4}	[-0.00962, 0.0107]	1.0000	
adaptive	pointer	0.1	-3.183×10^{-4}	[-0.0105, 0.00986]	1.0000	
adaptive	pointer	1	5.843×10^{-4}	[-0.00959, 0.0108]	1.0000	
adaptive	pointer	10	6.681×10^{-4}	[-0.00951, 0.0108]	1.0000	
adaptive	pointer	100	-3.246×10^{-4}	[-0.0105, 0.00985]	1.0000	
adaptive	pointer	1000	-0.004661	[-0.0148, 0.00551]	0.6834	
adaptive	pointer	1e+04	-0.003985	[-0.0142, 0.00619]	0.8066	
adaptive	seq2seq	0.01	0.003397	[-0.0179, 0.0247]	0.9979	
adaptive	seq2seq	0.1	0.003664	[-0.0177, 0.025]	0.9966	
adaptive	seq2seq	1	0.003076	[-0.0183, 0.0244]	0.9988	
adaptive	seq2seq	10	0.002487	[-0.0188, 0.0238]	0.9997	
adaptive	seq2seq	100	7.360×10^{-4}	[-0.0206, 0.0221]	1.0000	
adaptive	seq2seq	1000	0.008352	[-0.013, 0.0297]	0.8067	
adaptive	seq2seq	1e+04	0.008674	[-0.0127, 0.03]	0.7804	
tropical	pointer	0.01	-0.00556	[-0.0621, 0.051]	0.9999	
tropical	pointer	0.1	-0.004773	[-0.0613, 0.0517]	1.0000	
tropical	pointer	1	4.440×10^{-4}	[-0.0561, 0.057]	1.0000	
tropical	pointer	10	-0.001797	[-0.0583, 0.0547]	1.0000	
tropical	pointer	100	-0.003838	[-0.0603, 0.0527]	1.0000	
tropical	pointer	1000	-0.006183	[-0.0627, 0.0503]	0.9998	
tropical	pointer	1e+04	-0.007141	[-0.0637, 0.0494]	0.9995	
tropical	seq2seq	0.01	-0.003974	[-0.067, 0.059]	1.0000	
tropical	seq2seq	0.1	-0.001111	[-0.0641, 0.0619]	1.0000	
tropical	seq2seq	1	-0.004786	[-0.0678, 0.0582]	1.0000	
tropical	seq2seq	10	-0.003318	[-0.0663, 0.0597]	1.0000	
tropical	seq2seq	100	-0.003346	[-0.0663, 0.0596]	1.0000	
tropical	seq2seq	1000	-0.02216	[-0.0851, 0.0408]	0.8697	
tropical	seq2seq	1e+04	-0.08108	[-0.144, -0.0181]	0.0072	*
vanilla	pointer	0.01	0.00314	[-0.00384, 0.0101]	0.6998	
vanilla	pointer	0.1	0.003006	[-0.00398, 0.00999]	0.7366	
vanilla	pointer	1	0.003034	[-0.00395, 0.01]	0.7287	

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	10	0.002993	[-0.00399, 0.00998]	0.7400
vanilla	pointer	100	0.002383	[-0.0046, 0.00937]	0.8846
vanilla	pointer	1000	0.001711	[-0.00527, 0.00869]	0.9750
vanilla	pointer	1e+04	-1.009×10^{-4}	[-0.00708, 0.00688]	1.0000
vanilla	seq2seq	0.01	0.01141	[-0.0307, 0.0535]	0.9585
vanilla	seq2seq	0.1	0.01209	[-0.03, 0.0542]	0.9455
vanilla	seq2seq	1	0.01202	[-0.0301, 0.0541]	0.9469
vanilla	seq2seq	10	0.01757	[-0.0245, 0.0597]	0.7605
vanilla	seq2seq	100	0.006413	[-0.0357, 0.0485]	0.9984
vanilla	seq2seq	1000	-0.002442	[-0.0445, 0.0396]	1.0000
vanilla	seq2seq	1e+04	-0.03659	[-0.0787, 0.0055]	0.1096

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-8.295×10^{-4}	[-0.00126, -2.09×10^{-4}]	0.0850	-0.2	0.1316
adaptive	seq2seq	8.167×10^{-4}	[-0.00129, 0.00262]	0.4160	0.116	0.3945
tropical	pointer	-4.231×10^{-4}	[-0.00106, 2.16×10^{-4}]	0.8646	-0.0438	0.7461
tropical	seq2seq	-0.009714	[-0.0123, -0.0073]	0.0055	-0.33	0.0112
vanilla	pointer	-4.629×10^{-4}	[-6.89×10^{-4}, -2.37×10^{-4}]	0.1432	-0.215	0.1108
vanilla	seq2seq	-0.006381	[-0.0107, -0.00133]	0.0030	-0.478	0.0004

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.05881	[0.00771, 0.11]	0.1061	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.05011	[0.0138, 0.0865]	0.1061	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.05103	[0.0108, 0.0912]	0.1061	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.05622	[0.018, 0.0944]	0.1061	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.05402	[0.00858, 0.0995]	0.1061	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.05259	[0.017, 0.0882]	0.1061	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.05092	[0.0112, 0.0906]	0.1061	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.05177	[0.0179, 0.0857]	0.1061	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1665	[0.112, 0.221]	0.0125	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.1511	[0.0995, 0.203]	0.0125	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.1533	[0.104, 0.203]	0.0125	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1497	[0.102, 0.197]	0.0125	*

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1456	[0.109, 0.182]	0.0058	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1567	[0.118, 0.196]	0.0058	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.1468	[0.11, 0.184]	0.0058	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.122	[0.0719, 0.172]	0.0224	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.003014	[-0.0143, 0.00828]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.005594	[-0.0154, 0.00419]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.006338	[-0.0163, 0.00367]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.005464	[-0.0163, 0.0054]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.005338	[-0.0154, 0.00473]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.005721	[-0.0154, 0.004]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.009386	[-0.0173, -0.00149]	0.4190	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.006897	[-0.0193, 0.00546]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.01807	[-0.0644, 0.0283]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.02609	[-0.067, 0.0148]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.0265	[-0.0668, 0.0138]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.02702	[-0.0684, 0.0143]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.03316	[-0.0574, -0.00894]	0.3054	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.02375	[-0.0609, 0.0134]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00728	[-0.0417, 0.0272]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.02719	[0.0065, 0.0479]	0.3271	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.06473	[0.0488, 0.0807]	0.0075	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.0619	[0.044, 0.0798]	0.0113	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.06075	[0.0445, 0.077]	0.0098	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.06224	[0.0447, 0.0798]	0.0108	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.06292	[0.0459, 0.0799]	0.0098	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.06367	[0.0493, 0.078]	0.0057	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.05172	[0.0235, 0.0799]	0.0916	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.05208	[0.0425, 0.0617]	0.0027	*

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.058	[-0.0853, -0.0307]	0.0619	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.05958	[-0.107, -0.0126]	0.1860	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.06166	[-0.106, -0.0168]	0.1860	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.05277	[-0.0913, -0.0143]	0.1860	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.05648	[-0.0974, -0.0155]	0.1860	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.05849	[-0.0959, -0.0211]	0.1343	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.04202	[-0.0761, -0.0079]	0.1860	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.01594	[-0.0172, 0.0491]	0.2524	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.04967	[0.00313, 0.0962]	0.1860	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.0414	[0.00866, 0.0741]	0.1860	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.04059	[0.00794, 0.0732]	0.1860	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.04069	[0.00824, 0.0731]	0.1860	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.0351	[0.0199, 0.0503]	0.0485	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.04564	[0.0174, 0.0739]	0.1315	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.05383	[0.0276, 0.0801]	0.0657	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.08616	[0.0655, 0.107]	0.0070	*

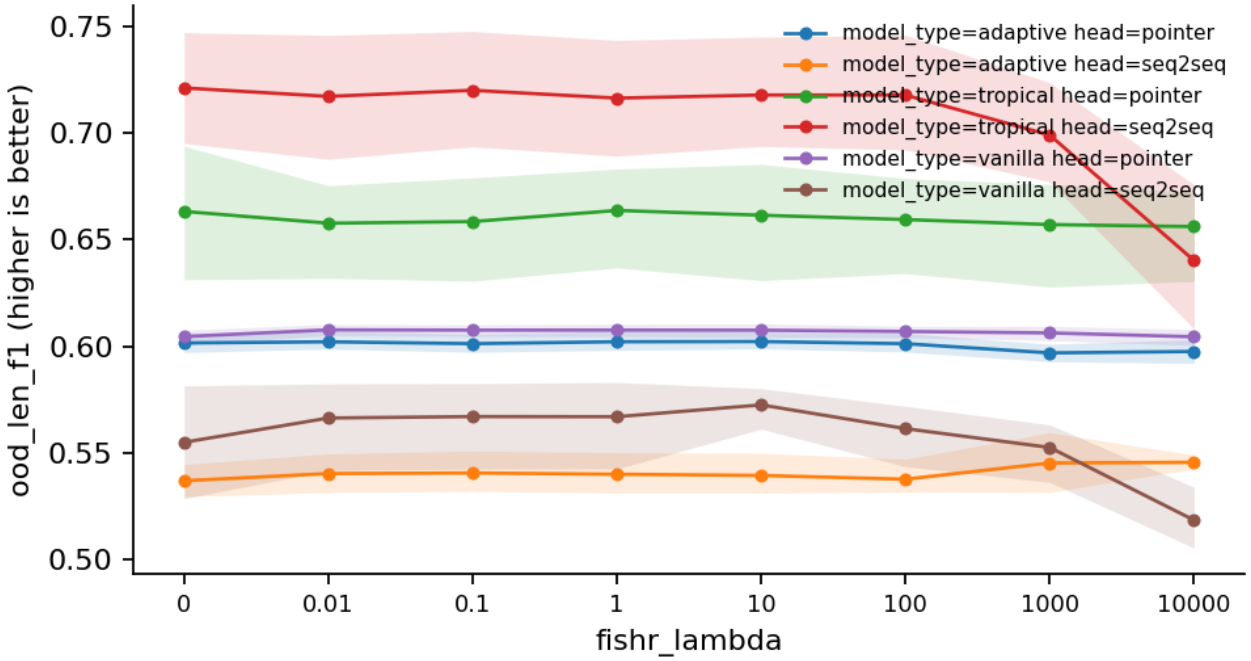


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

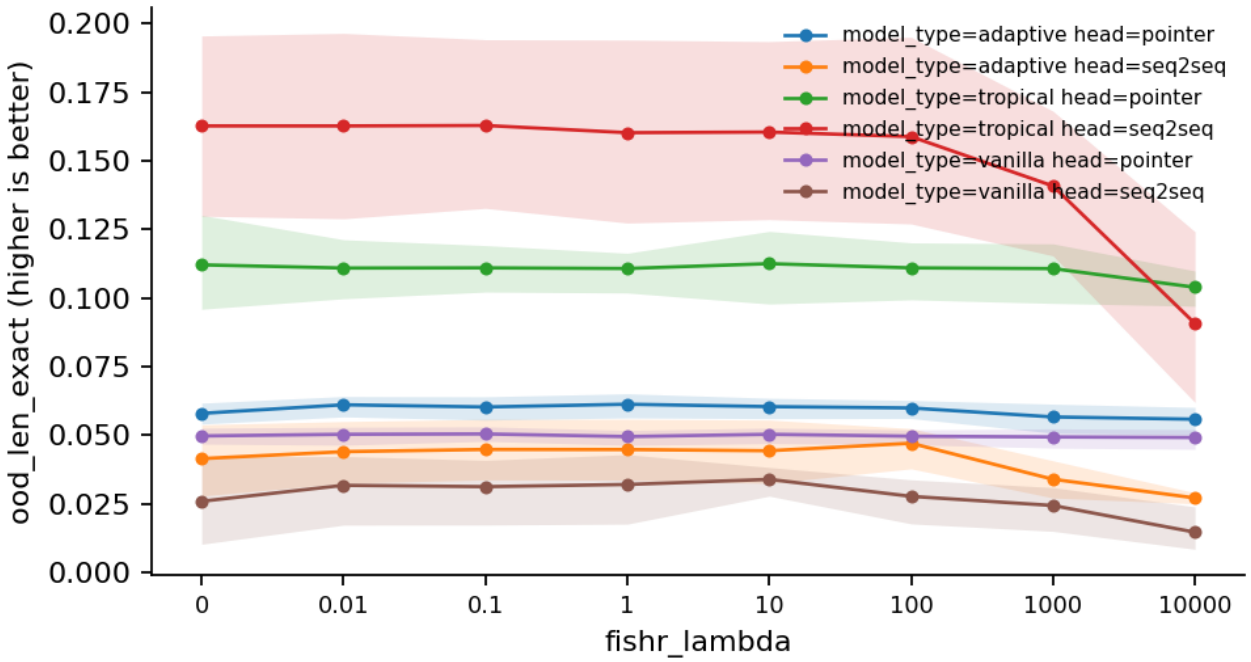


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.05793	[0.0541, 0.0617]	0.00455	5	0
adaptive	pointer	0.01	0.06107	[0.0567, 0.0641]	0.00502	5	0
adaptive	pointer	0.1	0.06027	[0.0555, 0.064]	0.00545	5	0
adaptive	pointer	1	0.06127	[0.0563, 0.0651]	0.00556	5	0
adaptive	pointer	10	0.0604	[0.0561, 0.0635]	0.00498	5	0
adaptive	pointer	100	0.0599	[0.0561, 0.0627]	0.00439	5	0
adaptive	pointer	1000	0.05663	[0.0508, 0.0613]	0.00689	5	0
adaptive	pointer	1e+04	0.0558	[0.0501, 0.0602]	0.00675	5	0
adaptive	seq2seq	0	0.04143	[0.0278, 0.0538]	0.0166	5	0
adaptive	seq2seq	0.01	0.04397	[0.0325, 0.055]	0.0148	5	0
adaptive	seq2seq	0.1	0.04477	[0.0337, 0.0558]	0.0148	5	0
adaptive	seq2seq	1	0.04473	[0.0336, 0.0558]	0.0148	5	0
adaptive	seq2seq	10	0.0443	[0.0325, 0.0555]	0.0154	5	0
adaptive	seq2seq	100	0.04703	[0.0378, 0.0524]	0.0103	5	0
adaptive	seq2seq	1000	0.0339	[0.0271, 0.0407]	0.00873	5	0
adaptive	seq2seq	1e+04	0.0271	[0.025, 0.029]	0.00269	5	0
tropical	pointer	0	0.1121	[0.096, 0.13]	0.0213	5	0
tropical	pointer	0.01	0.1109	[0.0998, 0.121]	0.0149	5	0
tropical	pointer	0.1	0.111	[0.102, 0.119]	0.0113	5	0
tropical	pointer	1	0.1108	[0.102, 0.116]	0.00985	5	0
tropical	pointer	10	0.1125	[0.0979, 0.124]	0.0176	5	0
tropical	pointer	100	0.111	[0.0994, 0.12]	0.0133	5	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	1000	0.1107	[0.0981, 0.12]	0.0142	5	0
tropical	pointer	1e+04	0.1039	[0.0971, 0.11]	0.00833	5	0
tropical	seq2seq	0	0.1628	[0.13, 0.196]	0.0418	5	0
tropical	seq2seq	0.01	0.1628	[0.129, 0.197]	0.044	5	0
tropical	seq2seq	0.1	0.1629	[0.133, 0.194]	0.0411	5	0
tropical	seq2seq	1	0.1603	[0.127, 0.194]	0.0451	5	0
tropical	seq2seq	10	0.1605	[0.129, 0.194]	0.0416	5	0
tropical	seq2seq	100	0.1589	[0.127, 0.195]	0.0432	5	0
tropical	seq2seq	1000	0.1409	[0.115, 0.168]	0.035	5	0
tropical	seq2seq	1e+04	0.09073	[0.062, 0.124]	0.039	5	0
vanilla	pointer	0	0.0497	[0.0468, 0.0526]	0.00374	5	0
vanilla	pointer	0.01	0.0503	[0.0464, 0.0529]	0.00437	5	0
vanilla	pointer	0.1	0.05043	[0.0478, 0.0531]	0.00357	5	0
vanilla	pointer	1	0.04947	[0.0464, 0.0517]	0.00362	5	0
vanilla	pointer	10	0.0503	[0.047, 0.0526]	0.00375	5	0
vanilla	pointer	100	0.0496	[0.0466, 0.0519]	0.00344	5	0
vanilla	pointer	1000	0.04933	[0.0453, 0.0523]	0.00439	5	0
vanilla	pointer	1e+04	0.04913	[0.0449, 0.0521]	0.00461	5	0
vanilla	seq2seq	0	0.02587	[0.0103, 0.042]	0.0206	5	0
vanilla	seq2seq	0.01	0.0317	[0.0172, 0.0423]	0.0164	5	0
vanilla	seq2seq	0.1	0.0312	[0.0173, 0.0408]	0.0157	5	0
vanilla	seq2seq	1	0.032	[0.0176, 0.0429]	0.017	5	0
vanilla	seq2seq	10	0.03387	[0.0278, 0.0382]	0.00699	5	0
vanilla	seq2seq	100	0.02767	[0.0177, 0.0337]	0.0112	5	0
vanilla	seq2seq	1000	0.02433	[0.015, 0.0309]	0.0104	5	0
vanilla	seq2seq	1e+04	0.0146	[0.00843, 0.0239]	0.00957	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.003133	[-0.00651, 0.0128]	0.9054
adaptive	pointer	0.1	0.002333	[-0.00731, 0.012]	0.9766
adaptive	pointer	1	0.003333	[-0.00631, 0.013]	0.8782
adaptive	pointer	10	0.002467	[-0.00717, 0.0121]	0.9688
adaptive	pointer	100	0.001967	[-0.00767, 0.0116]	0.9907
adaptive	pointer	1000	-0.0013	[-0.0109, 0.00834]	0.9992
adaptive	pointer	1e+04	-0.002133	[-0.0118, 0.00751]	0.9855
adaptive	seq2seq	0.01	0.002533	[-0.0202, 0.0253]	0.9998
adaptive	seq2seq	0.1	0.003333	[-0.0194, 0.0261]	0.9987
adaptive	seq2seq	1	0.0033	[-0.0194, 0.026]	0.9988
adaptive	seq2seq	10	0.002867	[-0.0199, 0.0256]	0.9995
adaptive	seq2seq	100	0.0056	[-0.0171, 0.0283]	0.9744
adaptive	seq2seq	1000	-0.007533	[-0.0303, 0.0152]	0.8978
adaptive	seq2seq	1e+04	-0.01433	[-0.0371, 0.00841]	0.3664
tropical	pointer	0.01	-0.001167	[-0.0263, 0.024]	1.0000
tropical	pointer	0.1	-0.0011	[-0.0262, 0.024]	1.0000
tropical	pointer	1	-0.001333	[-0.0265, 0.0238]	1.0000
tropical	pointer	10	4.333×10^{-4}	[-0.0247, 0.0256]	1.0000
tropical	pointer	100	-0.0011	[-0.0262, 0.024]	1.0000
tropical	pointer	1000	-0.001367	[-0.0265, 0.0238]	1.0000
tropical	pointer	1e+04	-0.008167	[-0.0333, 0.017]	0.9057
tropical	seq2seq	0.01	0	[-0.0724, 0.0724]	1.0000
tropical	seq2seq	0.1	1.667×10^{-4}	[-0.0722, 0.0726]	1.0000

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	seq2seq	1	-0.002467	[-0.0749, 0.0699]	1.0000
tropical	seq2seq	10	-0.002233	[-0.0746, 0.0702]	1.0000
tropical	seq2seq	100	-0.0039	[-0.0763, 0.0685]	1.0000
tropical	seq2seq	1000	-0.0219	[-0.0943, 0.0505]	0.9309
tropical	seq2seq	1e+04	-0.07203	[-0.144, 3.73×10^{-4}]	0.0517
vanilla	pointer	0.01	6.000×10^{-4}	[-0.00631, 0.00751]	1.0000
vanilla	pointer	0.1	7.333×10^{-4}	[-0.00618, 0.00765]	0.9998
vanilla	pointer	1	-2.333×10^{-4}	[-0.00715, 0.00668]	1.0000
vanilla	pointer	10	6.000×10^{-4}	[-0.00631, 0.00751]	1.0000
vanilla	pointer	100	-1.000×10^{-4}	[-0.00701, 0.00681]	1.0000
vanilla	pointer	1000	-3.667×10^{-4}	[-0.00728, 0.00655]	1.0000
vanilla	pointer	1e+04	-5.667×10^{-4}	[-0.00748, 0.00635]	1.0000
vanilla	seq2seq	0.01	0.005833	[-0.0189, 0.0306]	0.9796
vanilla	seq2seq	0.1	0.005333	[-0.0194, 0.0301]	0.9874
vanilla	seq2seq	1	0.006133	[-0.0186, 0.0309]	0.9735
vanilla	seq2seq	10	0.008	[-0.0167, 0.0327]	0.9077
vanilla	seq2seq	100	0.0018	[-0.0229, 0.0265]	1.0000
vanilla	seq2seq	1000	-0.001533	[-0.0263, 0.0232]	1.0000
vanilla	seq2seq	1e+04	-0.01127	[-0.036, 0.0135]	0.6890

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-8.726×10^{-4}	[-0.0012, -4.75×10^{-4}]	0.0603	-0.29	0.0284
adaptive	seq2seq	-0.002501	[-0.00441, -5.06×10^{-4}]	0.0231	-0.356	0.0078
tropical	pointer	-7.607×10^{-4}	[-0.00167, 2.61×10^{-4}]	0.4682	-0.133	0.3287
tropical	seq2seq	-0.009345	[-0.0112, -0.00751]	0.0118	-0.331	0.0114
vanilla	pointer	-1.988×10^{-4}	[-4.86×10^{-4}, 1.42×10^{-4}]	0.5278	-0.131	0.3335
vanilla	seq2seq	-0.002477	[-0.00479, 3.68×10^{-4}]	0.0258	-0.4	0.0022

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0624	[0.0377, 0.0871]	0.0133	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.06063	[0.0468, 0.0744]	0.0028	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.06057	[0.0493, 0.0718]	0.0016	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.0613	[0.0529, 0.0697]	0.0005	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.06223	[0.0442, 0.0803]	0.0067	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.0614	[0.0485, 0.0743]	0.0025	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.0614	[0.0482, 0.0746]	0.0025	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.0548	[0.0488, 0.0608]	0.0002	*

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1369	[0.0826, 0.191]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.1311	[0.0767, 0.185]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.1317	[0.0814, 0.182]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1283	[0.0726, 0.184]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1267	[0.0772, 0.176]	0.0133	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1312	[0.0834, 0.179]	0.0127	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.1165	[0.0762, 0.157]	0.0118	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.07613	[0.0266, 0.126]	0.0133	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.008233	[0.00429, 0.0122]	0.0439	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.01077	[0.00941, 0.0121]	0.0004	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.009833	[0.00735, 0.0123]	0.0047	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.0118	[0.00862, 0.015]	0.0055	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.0101	[0.008, 0.0122]	0.0024	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.0103	[0.00831, 0.0123]	0.0019	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.0073	[0.00319, 0.0114]	0.0627	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.006667	[0.00343, 0.00991]	0.0439	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.01557	[-0.0115, 0.0426]	0.5327	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.01227	[-0.00513, 0.0297]	0.5327	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.01357	[-0.00199, 0.0291]	0.4362	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.01273	[-0.0043, 0.0298]	0.5327	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01043	[-0.00365, 0.0245]	0.5327	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.01937	[0.0177, 0.0211]	$< 10^{-4}$	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.009567	[-0.00401, 0.0231]	0.5327	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.0125	[0.00334, 0.0217]	0.1349	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.0165	[-0.00253, 0.0355]	0.5792	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.0171	[0.00108, 0.0331]	0.5792	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0155	$[-8.76 \times 10^{-4}, 0.0319]$	0.5792	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.01653	$[3.61 \times 10^{-4}, 0.0327]$	0.5792	

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.0161	$[-4.84 \times 10^{-6}, 0.0322]$	0.5792	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.01287	[-0.00257, 0.0283]	0.5792	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.02273	[0.0094, 0.0361]	0.1814	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.0287	[0.0203, 0.0371]	0.0166	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.05067	[-0.0846, -0.0167]	0.2731	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.05183	$[-0.104, 5.94 \times 10^{-4}]$	0.5792	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.05193	[-0.102, -0.00167]	0.5792	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.04953	[-0.101, 0.00168]	0.5792	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.048	[-0.0917, -0.00434]	0.5691	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.04787	[-0.0928, -0.00291]	0.5792	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	-0.03013	[-0.0661, 0.00585]	0.5792	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.0132	[-0.0337, 0.0601]	0.5792	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.02383	[-0.00208, 0.0497]	0.5792	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.0186	[0.00312, 0.0341]	0.4923	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.01923	[0.0028, 0.0357]	0.5021	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.01747	$[-5.43 \times 10^{-5}, 0.035]$	0.5792	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.01643	[0.0111, 0.0218]	0.0234	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02193	[0.00598, 0.0379]	0.3389	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.025	[0.0108, 0.0392]	0.1695	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.03453	[0.0233, 0.0458]	0.0234	*

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8884	[0.879, 0.898]	0.0126	5	0
adaptive	pointer	0.01	0.8639	[0.843, 0.883]	0.0241	5	0
adaptive	pointer	0.1	0.8732	[0.858, 0.889]	0.02	5	0
adaptive	pointer	1	0.8738	[0.858, 0.889]	0.0199	5	0
adaptive	pointer	10	0.8746	[0.865, 0.885]	0.0132	5	0
adaptive	pointer	100	0.8744	[0.855, 0.893]	0.0243	5	0
adaptive	pointer	1000	0.8688	[0.843, 0.891]	0.0311	5	0
adaptive	pointer	1e+04	0.8599	[0.83, 0.89]	0.0396	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0	0.8962	[0.89, 0.902]	0.00793	5	0
adaptive	seq2seq	0.01	0.8903	[0.886, 0.895]	0.00631	5	0
adaptive	seq2seq	0.1	0.8918	[0.886, 0.898]	0.00797	5	0
adaptive	seq2seq	1	0.8903	[0.884, 0.899]	0.0097	5	0
adaptive	seq2seq	10	0.8957	[0.888, 0.903]	0.0105	5	0
adaptive	seq2seq	100	0.895	[0.889, 0.903]	0.00852	5	0
adaptive	seq2seq	1000	0.903	[0.9, 0.906]	0.0034	5	0
adaptive	seq2seq	1e+04	0.8897	[0.88, 0.897]	0.0106	5	0
tropical	pointer	0	0.8567	[0.85, 0.864]	0.00937	5	0
tropical	pointer	0.01	0.858	[0.827, 0.882]	0.036	5	0
tropical	pointer	0.1	0.8656	[0.855, 0.881]	0.0179	5	0
tropical	pointer	1	0.8737	[0.862, 0.884]	0.0133	5	0
tropical	pointer	10	0.8701	[0.86, 0.88]	0.0134	5	0
tropical	pointer	100	0.8654	[0.854, 0.877]	0.0149	5	0
tropical	pointer	1000	0.8639	[0.859, 0.869]	0.00631	5	0
tropical	pointer	1e+04	0.8572	[0.849, 0.865]	0.0106	5	0
tropical	seq2seq	0	0.8589	[0.844, 0.871]	0.0169	5	0
tropical	seq2seq	0.01	0.8612	[0.852, 0.87]	0.012	5	0
tropical	seq2seq	0.1	0.8705	[0.854, 0.89]	0.0221	5	0
tropical	seq2seq	1	0.8634	[0.854, 0.872]	0.0119	5	0
tropical	seq2seq	10	0.8686	[0.854, 0.883]	0.0186	5	0
tropical	seq2seq	100	0.8539	[0.844, 0.863]	0.0128	5	0
tropical	seq2seq	1000	0.8589	[0.839, 0.88]	0.0279	5	0
tropical	seq2seq	1e+04	0.8569	[0.84, 0.87]	0.0208	5	0
vanilla	pointer	0	0.9127	[0.903, 0.922]	0.0116	5	0
vanilla	pointer	0.01	0.9141	[0.91, 0.918]	0.00472	5	0
vanilla	pointer	0.1	0.9142	[0.91, 0.919]	0.00658	5	0
vanilla	pointer	1	0.9143	[0.91, 0.919]	0.00552	5	0
vanilla	pointer	10	0.9137	[0.909, 0.918]	0.00587	5	0
vanilla	pointer	100	0.9166	[0.912, 0.921]	0.00546	5	0
vanilla	pointer	1000	0.916	[0.911, 0.921]	0.00616	5	0
vanilla	pointer	1e+04	0.9129	[0.909, 0.918]	0.00536	5	0
vanilla	seq2seq	0	0.8759	[0.863, 0.887]	0.0154	5	0
vanilla	seq2seq	0.01	0.8764	[0.862, 0.886]	0.0164	5	0
vanilla	seq2seq	0.1	0.8806	[0.867, 0.891]	0.016	5	0
vanilla	seq2seq	1	0.8806	[0.87, 0.89]	0.0125	5	0
vanilla	seq2seq	10	0.8824	[0.88, 0.885]	0.00336	5	0
vanilla	seq2seq	100	0.8833	[0.876, 0.892]	0.0109	5	0
vanilla	seq2seq	1000	0.892	[0.888, 0.896]	0.00471	5	0
vanilla	seq2seq	1e+04	0.8803	[0.877, 0.885]	0.00528	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.02445	[-0.0674, 0.0185]	0.4698
adaptive	pointer	0.1	-0.01513	[-0.0581, 0.0278]	0.8688
adaptive	pointer	1	-0.01461	[-0.0576, 0.0283]	0.8859
adaptive	pointer	10	-0.01383	[-0.0568, 0.0291]	0.9092
adaptive	pointer	100	-0.01401	[-0.057, 0.0289]	0.9040
adaptive	pointer	1000	-0.01954	[-0.0625, 0.0234]	0.6891
adaptive	pointer	1e+04	-0.02844	[-0.0714, 0.0145]	0.3177
adaptive	seq2seq	0.01	-0.005887	[-0.0206, 0.00882]	0.7915
adaptive	seq2seq	0.1	-0.00432	[-0.019, 0.0104]	0.9395

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
adaptive	seq2seq	1	-0.005859	[-0.0206, 0.00885]	0.7948
adaptive	seq2seq	10	-4.641×10^{-4}	[-0.0152, 0.0142]	1.0000
adaptive	seq2seq	100	-0.001148	[-0.0159, 0.0136]	1.0000
adaptive	seq2seq	1000	0.006824	[-0.00788, 0.0215]	0.6719
adaptive	seq2seq	1e+04	-0.006506	[-0.0212, 0.0082]	0.7139
tropical	pointer	0.01	0.00138	[-0.0291, 0.0318]	1.0000
tropical	pointer	0.1	0.008931	[-0.0215, 0.0394]	0.9399
tropical	pointer	1	0.01703	[-0.0134, 0.0475]	0.4881
tropical	pointer	10	0.01347	[-0.017, 0.0439]	0.7136
tropical	pointer	100	0.00876	[-0.0217, 0.0392]	0.9450
tropical	pointer	1000	0.007287	[-0.0232, 0.0377]	0.9779
tropical	pointer	1e+04	5.125×10^{-4}	[-0.0299, 0.031]	1.0000
tropical	seq2seq	0.01	0.002364	[-0.0302, 0.0349]	1.0000
tropical	seq2seq	0.1	0.01168	[-0.0209, 0.0443]	0.8598
tropical	seq2seq	1	0.004544	[-0.028, 0.0371]	0.9991
tropical	seq2seq	10	0.009766	[-0.0228, 0.0423]	0.9336
tropical	seq2seq	100	-0.004994	[-0.0376, 0.0276]	0.9983
tropical	seq2seq	1000	8.736×10^{-5}	[-0.0325, 0.0327]	1.0000
tropical	seq2seq	1e+04	-0.001948	[-0.0345, 0.0306]	1.0000
vanilla	pointer	0.01	0.001394	[-0.0104, 0.0132]	0.9997
vanilla	pointer	0.1	0.001457	[-0.0103, 0.0132]	0.9996
vanilla	pointer	1	0.001548	[-0.0102, 0.0133]	0.9993
vanilla	pointer	10	9.674×10^{-4}	[-0.0108, 0.0127]	1.0000
vanilla	pointer	100	0.003885	[-0.00788, 0.0156]	0.8989
vanilla	pointer	1000	0.003289	[-0.00847, 0.0151]	0.9519
vanilla	pointer	1e+04	2.044×10^{-4}	[-0.0116, 0.012]	1.0000
vanilla	seq2seq	0.01	5.444×10^{-4}	[-0.0199, 0.021]	1.0000
vanilla	seq2seq	0.1	0.00477	[-0.0157, 0.0252]	0.9807
vanilla	seq2seq	1	0.004722	[-0.0157, 0.0252]	0.9818
vanilla	seq2seq	10	0.006483	[-0.014, 0.0269]	0.9151
vanilla	seq2seq	100	0.007404	[-0.0131, 0.0279]	0.8546
vanilla	seq2seq	1000	0.01616	[-0.0043, 0.0366]	0.1696
vanilla	seq2seq	1e+04	0.004416	[-0.016, 0.0249]	0.9873

Table 11: `ood_val_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-7.217×10^{-4}	[-0.00437, 0.00287]	0.7286	0.0895	0.5123
adaptive	seq2seq	8.979×10^{-4}	$[5.05 \times 10^{-4}$, 0.00148]	0.2380	0.189	0.1622
tropical	pointer	-5.058×10^{-4}	[-0.00316, 0.00263]	0.7382	-0.17	0.1988
tropical	seq2seq	-0.001631	[-0.00317, -5.83×10^{-4}]	0.2920	-0.0933	0.4883
vanilla	pointer	8.684×10^{-5}	$[-5.44 \times 10^{-4}$, 7.18×10^{-4}]	0.8499	0.0476	0.7291
vanilla	seq2seq	0.001324	$[-6.32 \times 10^{-4}$, 0.00418]	0.1588	0.0781	0.5685

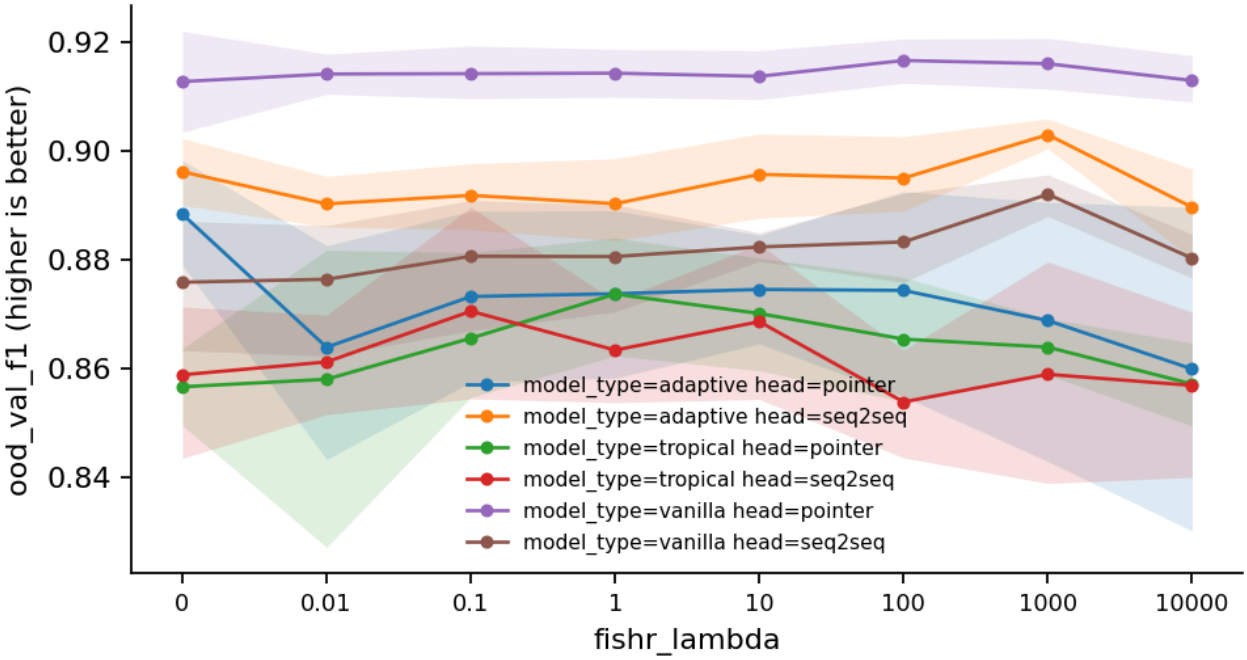


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.05608	[-0.0738, -0.0384]	0.0129	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.05609	[-0.0972, -0.015]	0.1739	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.04861	[-0.0733, -0.0239]	0.0599	
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.0406	[-0.0614, -0.0198]	0.0599	
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.04358	[-0.0577, -0.0294]	0.0134	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.0512	[-0.0735, -0.0289]	0.0371	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.05208	[-0.0603, -0.0439]	0.0010	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.05577	[-0.0709, -0.0407]	0.0077	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.01701	[-0.0443, 0.0103]	0.7952	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.01519	[-0.0465, 0.0162]	0.7952	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.0101	[-0.0565, 0.0363]	0.7952	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.01719	[-0.0455, 0.0111]	0.7952	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.01373	[-0.0367, 0.00923]	0.7952	

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.02941	[-0.0582, -6.52×10^{-4}]	0.3752	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.03308	[-0.0669, 7.82×10^{-4}]	0.3752	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.02337	[-0.0554, 0.00861]	0.6739	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.02436	[-0.0434, -0.00533]	0.2550	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.0502	[-0.0829, -0.0175]	0.1956	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.04095	[-0.0701, -0.0118]	0.2102	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.04052	[-0.0679, -0.0131]	0.2076	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.03915	[-0.056, -0.0223]	0.0471	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.04225	[-0.075, -0.00948]	0.2550	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.04719	[-0.0912, -0.0032]	0.3265	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.05301	[-0.105, -0.00122]	0.3265	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.02029	[0.00336, 0.0372]	0.2627	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.01386	[-0.00786, 0.0356]	0.6046	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.0112	[-0.0146, 0.037]	0.6046	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.009707	[-0.00107, 0.0205]	0.3334	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01334	[3.50×10^{-4}, 0.0263]	0.3265	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.01174	[-0.00842, 0.0319]	0.6046	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.01095	[0.00345, 0.0185]	0.2076	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.009367	[-0.00853, 0.0273]	0.6046	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.007783	[-0.0184, 0.00283]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.02634	[-0.0542, 0.0015]	0.8299	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.0186	[-0.0379, 6.82×10^{-4}]	0.8299	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.01654	[-0.0394, 0.00636]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.02114	[-0.0414, -8.87×10^{-4}]	0.7072	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.02064	[-0.0514, 0.0101]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.03415	[-0.0732, 0.00489]	0.9373	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.02972	[-0.0672, 0.00778]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.002203	[-0.0176, 0.0132]	1.0000	

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	-0.003187	[-0.06, 0.0536]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.004952	[-0.0362, 0.0263]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01028	[-0.00458, 0.0251]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.001502	[-0.0292, 0.0322]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01155	[-0.0123, 0.0354]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.004996	[-0.0305, 0.0405]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	2.573×10^{-4}	[-0.0283, 0.0289]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.03687	[0.0198, 0.0539]	0.0660	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.03772	[0.0218, 0.0536]	0.0525	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.03355	[0.0199, 0.0472]	0.0487	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.03369	[0.0244, 0.043]	0.0128	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.03135	[0.0207, 0.042]	0.0273	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.03335	[0.0188, 0.0479]	0.0568	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.024	[0.0146, 0.0334]	0.0437	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.03266	[0.0293, 0.0361]	0.0003	*

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.3602	[0.337, 0.385]	0.0312	5	0
adaptive	pointer	0.01	0.3127	[0.285, 0.345]	0.0387	5	0
adaptive	pointer	0.1	0.3266	[0.302, 0.352]	0.0337	5	0
adaptive	pointer	1	0.3321	[0.31, 0.355]	0.0298	5	0
adaptive	pointer	10	0.3294	[0.314, 0.35]	0.0229	5	0
adaptive	pointer	100	0.339	[0.309, 0.369]	0.0381	5	0
adaptive	pointer	1000	0.3262	[0.291, 0.357]	0.044	5	0
adaptive	pointer	1e+04	0.3176	[0.263, 0.372]	0.0681	5	0
adaptive	seq2seq	0	0.3653	[0.348, 0.384]	0.0234	5	0
adaptive	seq2seq	0.01	0.3514	[0.338, 0.364]	0.0178	5	0
adaptive	seq2seq	0.1	0.3553	[0.336, 0.374]	0.0242	5	0
adaptive	seq2seq	1	0.3504	[0.332, 0.375]	0.0278	5	0
adaptive	seq2seq	10	0.367	[0.342, 0.389]	0.0324	5	0
adaptive	seq2seq	100	0.3628	[0.344, 0.386]	0.0267	5	0
adaptive	seq2seq	1000	0.3816	[0.371, 0.394]	0.014	5	0
adaptive	seq2seq	1e+04	0.3424	[0.323, 0.359]	0.0235	5	0
tropical	pointer	0	0.2566	[0.235, 0.279]	0.0295	5	0
tropical	pointer	0.01	0.2869	[0.253, 0.321]	0.0425	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	pointer	0.1	0.286	[0.257, 0.328]	0.0459	5	0
tropical	pointer	1	0.2986	[0.266, 0.327]	0.0373	5	0
tropical	pointer	10	0.2898	[0.264, 0.317]	0.0331	5	0
tropical	pointer	100	0.2858	[0.251, 0.319]	0.044	5	0
tropical	pointer	1000	0.285	[0.263, 0.31]	0.0298	5	0
tropical	pointer	1e+04	0.2705	[0.258, 0.28]	0.0149	5	0
tropical	seq2seq	0	0.2669	[0.227, 0.298]	0.0441	5	0
tropical	seq2seq	0.01	0.2703	[0.249, 0.294]	0.0298	5	0
tropical	seq2seq	0.1	0.2923	[0.251, 0.334]	0.0551	5	0
tropical	seq2seq	1	0.2752	[0.248, 0.301]	0.0355	5	0
tropical	seq2seq	10	0.2873	[0.245, 0.325]	0.0513	5	0
tropical	seq2seq	100	0.2544	[0.231, 0.277]	0.0307	5	0
tropical	seq2seq	1000	0.2721	[0.218, 0.328]	0.0731	5	0
tropical	seq2seq	1e+04	0.2664	[0.227, 0.301]	0.0499	5	0
vanilla	pointer	0	0.4249	[0.393, 0.457]	0.0411	5	0
vanilla	pointer	0.01	0.4294	[0.412, 0.447]	0.0229	5	0
vanilla	pointer	0.1	0.4279	[0.407, 0.451]	0.0301	5	0
vanilla	pointer	1	0.4276	[0.409, 0.446]	0.0237	5	0
vanilla	pointer	10	0.4284	[0.41, 0.449]	0.0255	5	0
vanilla	pointer	100	0.438	[0.418, 0.454]	0.0235	5	0
vanilla	pointer	1000	0.4338	[0.417, 0.449]	0.0207	5	0
vanilla	pointer	1e+04	0.4258	[0.412, 0.439]	0.0178	5	0
vanilla	seq2seq	0	0.3047	[0.278, 0.328]	0.0326	5	0
vanilla	seq2seq	0.01	0.3061	[0.274, 0.33]	0.0363	5	0
vanilla	seq2seq	0.1	0.3171	[0.286, 0.342]	0.038	5	0
vanilla	seq2seq	1	0.3159	[0.294, 0.337]	0.0281	5	0
vanilla	seq2seq	10	0.319	[0.312, 0.327]	0.00961	5	0
vanilla	seq2seq	100	0.3211	[0.304, 0.34]	0.0237	5	0
vanilla	seq2seq	1000	0.336	[0.316, 0.356]	0.025	5	0
vanilla	seq2seq	1e+04	0.3123	[0.302, 0.324]	0.0139	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.0475	[-0.118, 0.023]	0.3024
adaptive	pointer	0.1	-0.03366	[-0.104, 0.0369]	0.6462
adaptive	pointer	1	-0.0281	[-0.0986, 0.0424]	0.7949
adaptive	pointer	10	-0.03088	[-0.101, 0.0397]	0.7226
adaptive	pointer	100	-0.0212	[-0.0917, 0.0493]	0.9328
adaptive	pointer	1000	-0.03404	[-0.105, 0.0365]	0.6355
adaptive	pointer	1e+04	-0.04268	[-0.113, 0.0279]	0.4074
adaptive	seq2seq	0.01	-0.01394	[-0.0565, 0.0286]	0.9019
adaptive	seq2seq	0.1	-0.01	[-0.0525, 0.0325]	0.9798
adaptive	seq2seq	1	-0.0149	[-0.0574, 0.0276]	0.8716
adaptive	seq2seq	10	0.0017	[-0.0408, 0.0442]	1.0000
adaptive	seq2seq	100	-0.00254	[-0.0451, 0.04]	1.0000
adaptive	seq2seq	1000	0.01624	[-0.0263, 0.0588]	0.8227
adaptive	seq2seq	1e+04	-0.0229	[-0.0654, 0.0196]	0.5266
tropical	pointer	0.01	0.0303	[-0.0324, 0.093]	0.6344
tropical	pointer	0.1	0.02938	[-0.0333, 0.0921]	0.6633
tropical	pointer	1	0.04198	[-0.0207, 0.105]	0.3080
tropical	pointer	10	0.03322	[-0.0295, 0.0959]	0.5437
tropical	pointer	100	0.0292	[-0.0335, 0.0919]	0.6687

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
tropical	pointer	1000	0.02844	[-0.0343, 0.0912]	0.6924
tropical	pointer	1e+04	0.0139	[-0.0488, 0.0766]	0.9854
tropical	seq2seq	0.01	0.0034	[-0.0807, 0.0875]	1.0000
tropical	seq2seq	0.1	0.02544	[-0.0586, 0.11]	0.9308
tropical	seq2seq	1	0.0083	[-0.0758, 0.0924]	0.9999
tropical	seq2seq	10	0.02038	[-0.0637, 0.104]	0.9764
tropical	seq2seq	100	-0.0125	[-0.0966, 0.0716]	0.9986
tropical	seq2seq	1000	0.00516	[-0.0789, 0.0892]	1.0000
tropical	seq2seq	1e+04	-4.800×10^{-4}	[-0.0846, 0.0836]	1.0000
vanilla	pointer	0.01	0.00452	[-0.0418, 0.0509]	0.9999
vanilla	pointer	0.1	0.00302	[-0.0433, 0.0494]	1.0000
vanilla	pointer	1	0.00272	[-0.0436, 0.0491]	1.0000
vanilla	pointer	10	0.00352	[-0.0428, 0.0499]	1.0000
vanilla	pointer	100	0.0131	[-0.0332, 0.0594]	0.9494
vanilla	pointer	1000	0.00888	[-0.0375, 0.0552]	0.9935
vanilla	pointer	1e+04	9.200×10^{-4}	[-0.0454, 0.0473]	1.0000
vanilla	seq2seq	0.01	0.00144	[-0.0468, 0.0497]	1.0000
vanilla	seq2seq	0.1	0.0124	[-0.0358, 0.0606]	0.9681
vanilla	seq2seq	1	0.01126	[-0.037, 0.0595]	0.9806
vanilla	seq2seq	10	0.01438	[-0.0338, 0.0626]	0.9351
vanilla	seq2seq	100	0.01648	[-0.0317, 0.0647]	0.8837
vanilla	seq2seq	1000	0.0313	[-0.0169, 0.0795]	0.3372
vanilla	seq2seq	1e+04	0.00764	[-0.0406, 0.0559]	0.9979

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	7.357×10^{-4}	[-0.00669, 0.00843]	0.8254	0.103	0.4461
adaptive	seq2seq	0.001356	$[-7.58 \times 10^{-4}$, 0.00347]	0.5343	0.107	0.4159
tropical	pointer	-0.002281	[-0.00805, 0.00366]	0.4387	-0.08	0.5449
tropical	seq2seq	-0.002607	[-0.00599, 0.00119]	0.5085	-0.0705	0.6107
vanilla	pointer	4.036×10^{-4}	[-0.00165, 0.0031]	0.8312	0.0362	0.7808
vanilla	seq2seq	0.002201	[-0.00291, 0.00989]	0.3207	0.0267	0.8556

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.1683	[-0.223, -0.114]	0.0144	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	-0.1426	[-0.189, -0.0962]	0.0144	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	-0.142	[-0.221, -0.063]	0.0677	
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.1291	[-0.196, -0.0619]	0.0607	
model_type: tropical - vanilla head=pointer fishr_lambda=10	-0.1386	[-0.173, -0.104]	0.0054	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	-0.1522	[-0.23, -0.0747]	0.0607	

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	-0.1488	[-0.203, -0.0941]	0.0198	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	-0.1554	[-0.189, -0.122]	0.0032	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.03776	[-0.109, 0.0335]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.0358	[-0.104, 0.0327]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.02472	[-0.137, 0.0873]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	-0.04072	[-0.117, 0.0358]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.03176	[-0.0944, 0.0308]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.06674	[-0.129, -0.00398]	0.3349	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.0639	[-0.143, 0.0155]	0.6251	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-0.04588	[-0.123, 0.0317]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.06468	[-0.124, -0.00489]	0.2582	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.1167	[-0.179, -0.0546]	0.0964	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.1014	[-0.169, -0.0335]	0.1719	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.0955	[-0.156, -0.0354]	0.1623	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.09908	[-0.139, -0.0591]	0.0373	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.09898	[-0.167, -0.0305]	0.1751	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.1076	[-0.183, -0.0324]	0.1751	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.1083	[-0.191, -0.0256]	0.1987	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.06066	[0.0221, 0.0992]	0.1623	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.04528	[-0.00867, 0.0992]	0.3209	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.03826	[-0.0301, 0.107]	0.3209	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.0345	[0.00212, 0.0669]	0.2582	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.04798	[0.00997, 0.086]	0.1987	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.04164	[-0.013, 0.0963]	0.3209	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.0456	[0.00451, 0.0867]	0.2582	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.03012	[-0.0117, 0.072]	0.3209	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.00508	[-0.0394, 0.0293]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.03864	[-0.0781, 8.02×10^{-4}]	0.7948	

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.02874	[-0.0459, -0.0116]	0.1537	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.01828	[-0.0614, 0.0248]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.03766	[-0.0784, 0.00308]	0.8710	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.02374	[-0.0803, 0.0328]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.05536	[-0.116, 0.00558]	0.8710	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.02486	[-0.0886, 0.0388]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.01032	[-0.0475, 0.0269]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.01658	[-0.0708, 0.104]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	-0.00638	[-0.0637, 0.051]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.02336	[-0.0213, 0.0681]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.00252	[-0.0815, 0.0866]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.03138	[-0.0259, 0.0887]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.01296	[-0.0604, 0.0863]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.00406	[-0.046, 0.0541]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.1203	[0.0762, 0.164]	0.0324	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1233	[0.0972, 0.149]	0.0041	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1109	[0.0884, 0.133]	0.0036	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1117	[0.102, 0.121]	0.0001	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1094	[0.0692, 0.15]	0.0324	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1169	[0.074, 0.16]	0.0324	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.09784	[0.0488, 0.147]	0.0883	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1135	[0.0975, 0.13]	0.0009	*

A.4 `ood_noise_f1`

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8602	[0.857, 0.864]	0.0049	5	0
adaptive	pointer	0.01	0.8599	[0.856, 0.864]	0.00487	5	0
adaptive	pointer	0.1	0.8604	[0.856, 0.865]	0.00536	5	0
adaptive	pointer	1	0.8595	[0.855, 0.864]	0.00575	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	10	0.8591	[0.855, 0.864]	0.00562	5	0
adaptive	pointer	100	0.8606	[0.856, 0.865]	0.00542	5	0
adaptive	pointer	1000	0.8602	[0.857, 0.864]	0.00458	5	0
adaptive	pointer	1e+04	0.8619	[0.859, 0.865]	0.00392	5	0
adaptive	seq2seq	0	0.8461	[0.841, 0.851]	0.00663	5	0
adaptive	seq2seq	0.01	0.8495	[0.843, 0.856]	0.00779	5	0
adaptive	seq2seq	0.1	0.8489	[0.842, 0.856]	0.0086	5	0
adaptive	seq2seq	1	0.8489	[0.843, 0.856]	0.0086	5	0
adaptive	seq2seq	10	0.8481	[0.841, 0.855]	0.00834	5	0
adaptive	seq2seq	100	0.8517	[0.847, 0.855]	0.00545	5	0
adaptive	seq2seq	1000	0.8426	[0.84, 0.846]	0.00384	5	0
adaptive	seq2seq	1e+04	0.8181	[0.811, 0.826]	0.00954	5	0
tropical	pointer	0	0.869	[0.866, 0.872]	0.00394	5	0
tropical	pointer	0.01	0.8689	[0.865, 0.873]	0.00524	5	0
tropical	pointer	0.1	0.8686	[0.864, 0.873]	0.00545	5	0
tropical	pointer	1	0.8689	[0.863, 0.875]	0.00839	5	0
tropical	pointer	10	0.869	[0.865, 0.874]	0.00609	5	0
tropical	pointer	100	0.8707	[0.868, 0.874]	0.00401	5	0
tropical	pointer	1000	0.8676	[0.866, 0.87]	0.00275	5	0
tropical	pointer	1e+04	0.8681	[0.863, 0.873]	0.00593	5	0
tropical	seq2seq	0	0.8572	[0.851, 0.863]	0.00791	5	0
tropical	seq2seq	0.01	0.8563	[0.846, 0.867]	0.0135	5	0
tropical	seq2seq	0.1	0.8539	[0.843, 0.864]	0.0134	5	0
tropical	seq2seq	1	0.8613	[0.855, 0.867]	0.00774	5	0
tropical	seq2seq	10	0.8594	[0.852, 0.866]	0.00916	5	0
tropical	seq2seq	100	0.8592	[0.85, 0.866]	0.0109	5	0
tropical	seq2seq	1000	0.8605	[0.852, 0.869]	0.011	5	0
tropical	seq2seq	1e+04	0.8596	[0.853, 0.867]	0.00946	5	0
vanilla	pointer	0	0.8593	[0.857, 0.861]	0.00247	5	0
vanilla	pointer	0.01	0.8601	[0.859, 0.861]	0.00177	5	0
vanilla	pointer	0.1	0.86	[0.859, 0.861]	0.00138	5	0
vanilla	pointer	1	0.8596	[0.858, 0.861]	0.00169	5	0
vanilla	pointer	10	0.8598	[0.858, 0.861]	0.00207	5	0
vanilla	pointer	100	0.8601	[0.859, 0.861]	0.0015	5	0
vanilla	pointer	1000	0.8613	[0.861, 0.862]	0.00103	5	0
vanilla	pointer	1e+04	0.8611	[0.86, 0.862]	0.00109	5	0
vanilla	seq2seq	0	0.8565	[0.854, 0.859]	0.00345	5	0
vanilla	seq2seq	0.01	0.8553	[0.851, 0.859]	0.00523	5	0
vanilla	seq2seq	0.1	0.8548	[0.85, 0.859]	0.00541	5	0
vanilla	seq2seq	1	0.8544	[0.85, 0.858]	0.00496	5	0
vanilla	seq2seq	10	0.8538	[0.851, 0.857]	0.00397	5	0
vanilla	seq2seq	100	0.85	[0.846, 0.855]	0.00606	5	0
vanilla	seq2seq	1000	0.845	[0.839, 0.852]	0.00853	5	0
vanilla	seq2seq	1e+04	0.8378	[0.83, 0.846]	0.01	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-2.568×10^{-4}	[-0.00914, 0.00862]	1.0000
adaptive	pointer	0.1	2.143×10^{-4}	[-0.00867, 0.0091]	1.0000
adaptive	pointer	1	-6.699×10^{-4}	[-0.00955, 0.00821]	1.0000
adaptive	pointer	10	-0.001082	[-0.00996, 0.0078]	0.9996
adaptive	pointer	100	4.157×10^{-4}	[-0.00847, 0.0093]	1.0000

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	1000	2.933×10^{-5}	[-0.00885, 0.00891]	1.0000	
adaptive	pointer	1e+04	0.001723	[-0.00716, 0.0106]	0.9931	
adaptive	seq2seq	0.01	0.003448	[-0.00976, 0.0167]	0.9656	
adaptive	seq2seq	0.1	0.002857	[-0.0104, 0.0161]	0.9872	
adaptive	seq2seq	1	0.002852	[-0.0104, 0.0161]	0.9873	
adaptive	seq2seq	10	0.002012	[-0.0112, 0.0152]	0.9984	
adaptive	seq2seq	100	0.005621	[-0.00759, 0.0188]	0.7456	
adaptive	seq2seq	1000	-0.003458	[-0.0167, 0.00975]	0.9651	
adaptive	seq2seq	1e+04	-0.02792	[-0.0411, -0.0147]	$< 10^{-4}$	*
tropical	pointer	0.01	-7.411×10^{-5}	[-0.00962, 0.00947]	1.0000	
tropical	pointer	0.1	-3.733×10^{-4}	[-0.00992, 0.00917]	1.0000	
tropical	pointer	1	-1.089×10^{-4}	[-0.00965, 0.00944]	1.0000	
tropical	pointer	10	-2.563×10^{-5}	[-0.00957, 0.00952]	1.0000	
tropical	pointer	100	0.001674	[-0.00787, 0.0112]	0.9962	
tropical	pointer	1000	-0.001438	[-0.011, 0.00811]	0.9985	
tropical	pointer	1e+04	-9.146×10^{-4}	[-0.0105, 0.00863]	0.9999	
tropical	seq2seq	0.01	-9.374×10^{-4}	[-0.0195, 0.0176]	1.0000	
tropical	seq2seq	0.1	-0.00335	[-0.0219, 0.0152]	0.9954	
tropical	seq2seq	1	0.004111	[-0.0144, 0.0226]	0.9852	
tropical	seq2seq	10	0.002193	[-0.0163, 0.0207]	0.9997	
tropical	seq2seq	100	0.002015	[-0.0165, 0.0205]	0.9998	
tropical	seq2seq	1000	0.003296	[-0.0152, 0.0218]	0.9958	
tropical	seq2seq	1e+04	0.002406	[-0.0161, 0.0209]	0.9994	
vanilla	pointer	0.01	7.460×10^{-4}	[-0.0022, 0.00369]	0.9706	
vanilla	pointer	0.1	6.349×10^{-4}	[-0.00231, 0.00358]	0.9875	
vanilla	pointer	1	2.582×10^{-4}	[-0.00269, 0.00321]	1.0000	
vanilla	pointer	10	4.334×10^{-4}	[-0.00252, 0.00338]	0.9987	
vanilla	pointer	100	7.489×10^{-4}	[-0.0022, 0.0037]	0.9700	
vanilla	pointer	1000	0.00197	$[-9.79 \times 10^{-4}$, 0.00492]	0.3098	
vanilla	pointer	1e+04	0.00172	[-0.00123, 0.00467]	0.4448	
vanilla	seq2seq	0.01	-0.001154	[-0.0122, 0.00988]	0.9999	
vanilla	seq2seq	0.1	-0.001621	[-0.0127, 0.00941]	0.9987	
vanilla	seq2seq	1	-0.002078	[-0.0131, 0.00896]	0.9942	
vanilla	seq2seq	10	-0.002708	[-0.0137, 0.00833]	0.9748	
vanilla	seq2seq	100	-0.006512	[-0.0175, 0.00452]	0.4329	
vanilla	seq2seq	1000	-0.0115	$[-0.0225, -4.68 \times 10^{-4}]$	0.0381	*
vanilla	seq2seq	1e+04	-0.01868	[-0.0297, -0.00764]	0.0003	*

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	2.377×10^{-4}	$[4.38 \times 10^{-5}, 3.89 \times 10^{-4}]$	0.5585	0.0857	0.5355
adaptive	seq2seq	-0.003713	[-0.00494, -0.00282]	0.0003	-0.406	0.0028
tropical	pointer	-1.024×10^{-4}	$[-0.00118, 8.13 \times 10^{-4}]$	0.8199	-0.0324	0.8260
tropical	seq2seq	7.582×10^{-4}	$[-5.37 \times 10^{-5}, 0.00158]$	0.3879	0.0895	0.5149
vanilla	pointer	2.172×10^{-4}	$[1.01 \times 10^{-4}, 3.79 \times 10^{-4}]$	0.0930	0.219	0.1008
vanilla	seq2seq	-0.002742	[-0.00384, -0.00206]	$< 10^{-4}$	-0.505	0.0004

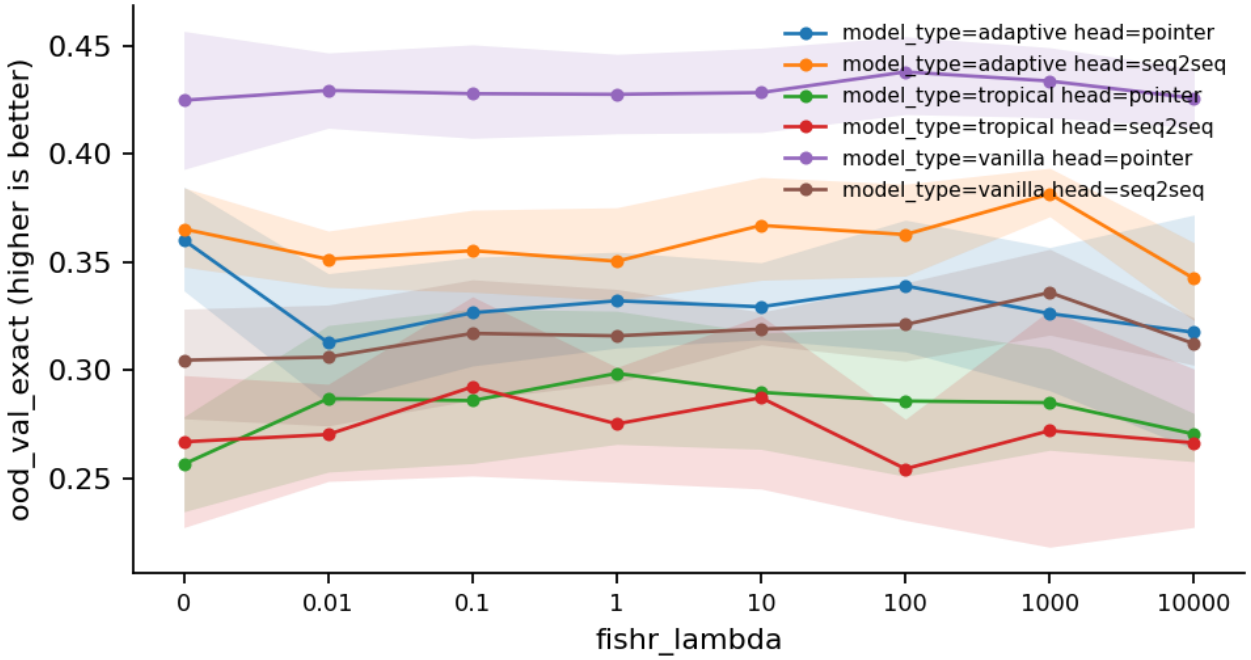


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

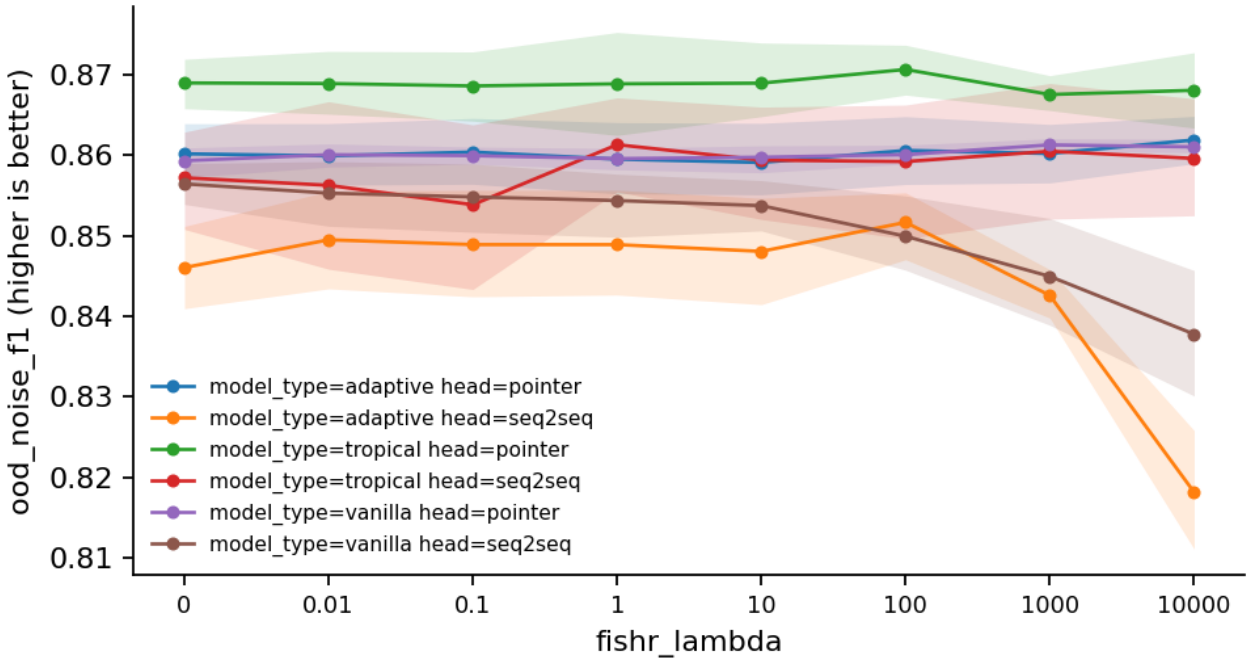


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.00967	[0.00215, 0.0172]	0.3039	
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.00885	[5.81×10^{-4}, 0.0171]	0.4518	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.008662	[3.25×10^{-4}, 0.017]	0.4518	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.009303	[-0.00275, 0.0214]	0.7685	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.009211	[0.00162, 0.0168]	0.3372	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.0106	[0.00397, 0.0172]	0.1585	
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.006263	[0.0026, 0.00993]	0.1352	
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.007036	[1.61×10^{-4}, 0.0139]	0.4518	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	7.552×10^{-4}	[-0.0125, 0.014]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	9.717×10^{-4}	[-0.0154, 0.0174]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-9.738×10^{-4}	[-0.0187, 0.0167]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.006944	[-0.00285, 0.0167]	0.7685	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.005656	[-0.00233, 0.0136]	0.7685	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.009282	[-0.00384, 0.0224]	0.7685	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.01556	[-0.00437, 0.0355]	0.7685	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.02184	[0.0124, 0.0312]	0.0476	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	8.617×10^{-4}	[-0.00546, 0.00719]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-1.410×10^{-4}	[-0.00606, 0.00578]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	4.411×10^{-4}	[-0.00622, 0.0071]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-6.638×10^{-5}	[-0.00749, 0.00736]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-6.535×10^{-4}	[-0.00757, 0.00626]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	5.285×10^{-4}	[-0.0067, 0.00776]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.001079	[-0.00706, 0.0049]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	8.648×10^{-4}	[-0.00396, 0.00569]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.01041	[-0.0213, 5.21×10^{-4}]	0.8600	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.005805	[-0.0206, 0.00901]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.005928	[-0.022, 0.0101]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.005476	[-0.0211, 0.0101]	1.0000	

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.005686	[-0.0161, 0.00469]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.001727	[-0.0126, 0.016]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00236	[-0.0171, 0.0123]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.01965	[-0.0212, -0.0181]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.01414	[0.00686, 0.0214]	0.1199	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.01044	$[-7.00 \times 10^{-4}$, 0.0216]	0.8479	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0115	$[-8.89 \times 10^{-4}$, 0.0239]	0.8479	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.01062	[-0.0033, 0.0245]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.01105	[-0.00278, 0.0249]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.008936	$[2.30 \times 10^{-4}$, 0.0176]	0.7889	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01763	[0.0134, 0.0218]	0.0074	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.04378	[0.0314, 0.0562]	0.0139	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.01179	[0.00387, 0.0197]	0.2892	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.01265	[-0.0075, 0.0328]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.01477	[-0.00335, 0.0329]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.007569	[-0.00771, 0.0229]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.00957	[-0.00714, 0.0263]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01145	$[-4.69 \times 10^{-4}$, 0.0234]	0.8479	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.007055	[-0.00712, 0.0212]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.008468	[-0.00422, 0.0212]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.002874	[-0.00338, 0.00912]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.004774	[-0.00267, 0.0122]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.00513	[-0.00246, 0.0127]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.00521	[-0.00176, 0.0122]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.006015	$[-1.25 \times 10^{-4}$, 0.0122]	0.8479	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.01013	$[8.98 \times 10^{-4}$, 0.0194]	0.6869	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01635	[0.0051, 0.0276]	0.2975	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.02327	[0.0121, 0.0345]	0.0990	

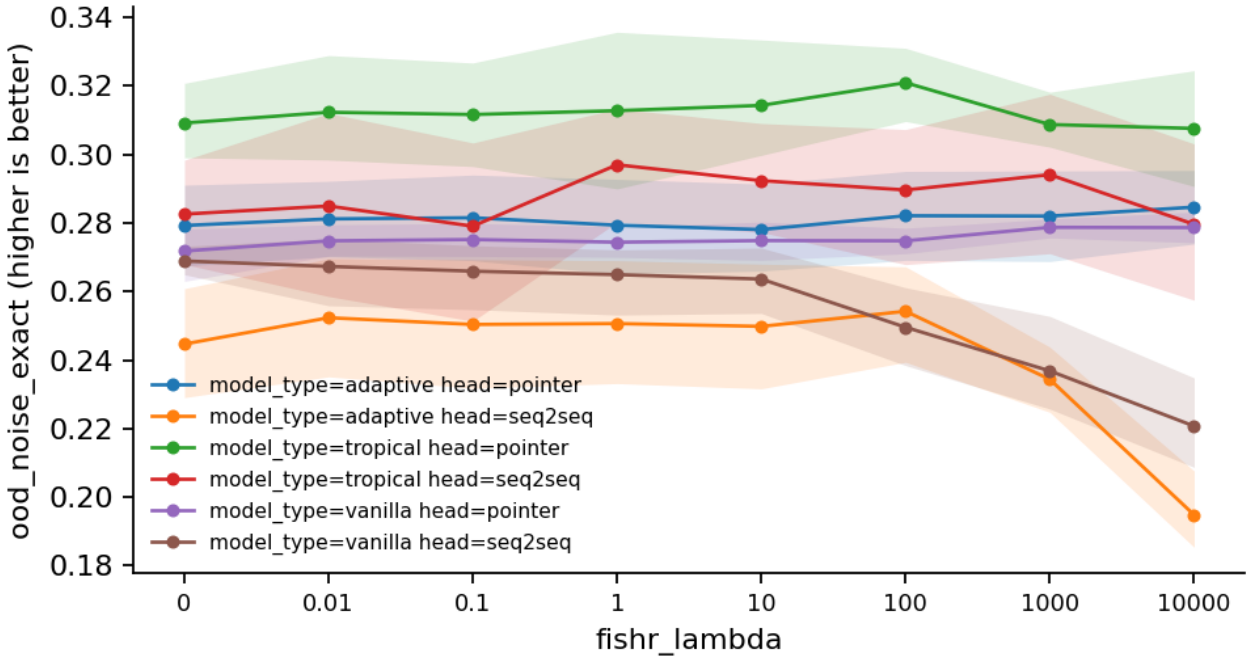


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.2793	[0.267, 0.291]	0.0161	5	0
adaptive	pointer	0.01	0.2812	[0.27, 0.292]	0.0143	5	0
adaptive	pointer	0.1	0.2815	[0.269, 0.294]	0.0165	5	0
adaptive	pointer	1	0.2793	[0.265, 0.293]	0.018	5	0
adaptive	pointer	10	0.2781	[0.266, 0.291]	0.0162	5	0
adaptive	pointer	100	0.2821	[0.269, 0.295]	0.0166	5	0
adaptive	pointer	1000	0.282	[0.269, 0.295]	0.0169	5	0
adaptive	pointer	1e+04	0.2846	[0.274, 0.295]	0.0142	5	0
adaptive	seq2seq	0	0.2447	[0.229, 0.261]	0.0203	5	0
adaptive	seq2seq	0.01	0.2523	[0.235, 0.27]	0.0223	5	0
adaptive	seq2seq	0.1	0.2504	[0.232, 0.269]	0.0245	5	0
adaptive	seq2seq	1	0.2506	[0.233, 0.269]	0.0239	5	0
adaptive	seq2seq	10	0.2498	[0.232, 0.268]	0.0234	5	0
adaptive	seq2seq	100	0.2542	[0.239, 0.267]	0.0188	5	0
adaptive	seq2seq	1000	0.2344	[0.225, 0.244]	0.0123	5	0
adaptive	seq2seq	1e+04	0.1948	[0.185, 0.208]	0.0147	5	0
tropical	pointer	0	0.3092	[0.299, 0.321]	0.013	5	0
tropical	pointer	0.01	0.3123	[0.298, 0.329]	0.0199	5	0
tropical	pointer	0.1	0.3116	[0.296, 0.327]	0.0191	5	0
tropical	pointer	1	0.3128	[0.29, 0.336]	0.0297	5	0
tropical	pointer	10	0.3143	[0.3, 0.333]	0.0225	5	0
tropical	pointer	100	0.3209	[0.31, 0.331]	0.0138	5	0
tropical	pointer	1000	0.3087	[0.302, 0.318]	0.011	5	0
tropical	pointer	1e+04	0.3076	[0.291, 0.324]	0.0216	5	0
tropical	seq2seq	0	0.2826	[0.268, 0.298]	0.0206	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0.01	0.2849	[0.259, 0.312]	0.034	5	0
tropical	seq2seq	0.1	0.2791	[0.251, 0.303]	0.0331	5	0
tropical	seq2seq	1	0.297	[0.28, 0.313]	0.0214	5	0
tropical	seq2seq	10	0.2923	[0.277, 0.309]	0.0215	5	0
tropical	seq2seq	100	0.2896	[0.268, 0.307]	0.0255	5	0
tropical	seq2seq	1000	0.2941	[0.271, 0.318]	0.0306	5	0
tropical	seq2seq	1e+04	0.2796	[0.258, 0.303]	0.0293	5	0
vanilla	pointer	0	0.2718	[0.263, 0.278]	0.0102	5	0
vanilla	pointer	0.01	0.2748	[0.27, 0.279]	0.00601	5	0
vanilla	pointer	0.1	0.2752	[0.27, 0.28]	0.00617	5	0
vanilla	pointer	1	0.2744	[0.27, 0.279]	0.00566	5	0
vanilla	pointer	10	0.2748	[0.269, 0.28]	0.00732	5	0
vanilla	pointer	100	0.2748	[0.271, 0.278]	0.00504	5	0
vanilla	pointer	1000	0.2787	[0.276, 0.281]	0.00358	5	0
vanilla	pointer	1e+04	0.2786	[0.274, 0.283]	0.00582	5	0
vanilla	seq2seq	0	0.2689	[0.265, 0.273]	0.00558	5	0
vanilla	seq2seq	0.01	0.2673	[0.256, 0.275]	0.0127	5	0
vanilla	seq2seq	0.1	0.2659	[0.255, 0.273]	0.0124	5	0
vanilla	seq2seq	1	0.2649	[0.253, 0.272]	0.0132	5	0
vanilla	seq2seq	10	0.2636	[0.254, 0.273]	0.0123	5	0
vanilla	seq2seq	100	0.2496	[0.238, 0.261]	0.015	5	0
vanilla	seq2seq	1000	0.2368	[0.226, 0.253]	0.018	5	0
vanilla	seq2seq	1e+04	0.2207	[0.209, 0.235]	0.0171	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.00192	[-0.0263, 0.0301]	1.0000	
adaptive	pointer	0.1	0.00226	[-0.026, 0.0305]	1.0000	
adaptive	pointer	1	8.000×10^{-5}	[-0.0281, 0.0283]	1.0000	
adaptive	pointer	10	-0.0012	[-0.0294, 0.027]	1.0000	
adaptive	pointer	100	0.00284	[-0.0254, 0.0311]	0.9999	
adaptive	pointer	1000	0.00274	[-0.0255, 0.031]	0.9999	
adaptive	pointer	1e+04	0.00536	[-0.0229, 0.0336]	0.9939	
adaptive	seq2seq	0.01	0.00768	[-0.0281, 0.0434]	0.9876	
adaptive	seq2seq	0.1	0.00572	[-0.03, 0.0415]	0.9978	
adaptive	seq2seq	1	0.00598	[-0.0298, 0.0417]	0.9971	
adaptive	seq2seq	10	0.00516	[-0.0306, 0.0409]	0.9988	
adaptive	seq2seq	100	0.00956	[-0.0262, 0.0453]	0.9612	
adaptive	seq2seq	1000	-0.0103	[-0.046, 0.0254]	0.9446	
adaptive	seq2seq	1e+04	-0.04984	[-0.0856, -0.0141]	0.0033	*
tropical	pointer	0.01	0.00314	[-0.0312, 0.0375]	0.9999	
tropical	pointer	0.1	0.00248	[-0.0319, 0.0368]	1.0000	
tropical	pointer	1	0.0036	[-0.0307, 0.0379]	0.9999	
tropical	pointer	10	0.00514	[-0.0292, 0.0395]	0.9985	
tropical	pointer	100	0.01176	[-0.0226, 0.0461]	0.8829	
tropical	pointer	1000	-4.400×10^{-4}	[-0.0348, 0.0339]	1.0000	
tropical	pointer	1e+04	-0.00158	[-0.0359, 0.0328]	1.0000	
tropical	seq2seq	0.01	0.00236	[-0.0457, 0.0504]	1.0000	
tropical	seq2seq	0.1	-0.00352	[-0.0515, 0.0445]	1.0000	
tropical	seq2seq	1	0.0144	[-0.0336, 0.0624]	0.9335	
tropical	seq2seq	10	0.00976	[-0.0383, 0.0578]	0.9909	
tropical	seq2seq	100	0.00704	[-0.041, 0.0551]	0.9987	

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
tropical	seq2seq	1000	0.0115	[-0.0365, 0.0595]	0.9778	
tropical	seq2seq	1e+04	-0.00298	[-0.051, 0.045]	1.0000	
vanilla	pointer	0.01	0.00298	[-0.00835, 0.0143]	0.9643	
vanilla	pointer	0.1	0.00336	[-0.00797, 0.0147]	0.9369	
vanilla	pointer	1	0.00258	[-0.00875, 0.0139]	0.9831	
vanilla	pointer	10	0.00304	[-0.00829, 0.0144]	0.9606	
vanilla	pointer	100	0.00298	[-0.00835, 0.0143]	0.9643	
vanilla	pointer	1000	0.00692	[-0.00441, 0.0183]	0.3980	
vanilla	pointer	1e+04	0.00684	[-0.00449, 0.0182]	0.4098	
vanilla	seq2seq	0.01	-0.00158	[-0.0256, 0.0224]	1.0000	
vanilla	seq2seq	0.1	-0.00298	[-0.027, 0.021]	0.9996	
vanilla	seq2seq	1	-0.00398	[-0.028, 0.02]	0.9973	
vanilla	seq2seq	10	-0.00528	[-0.0293, 0.0187]	0.9860	
vanilla	seq2seq	100	-0.01934	[-0.0434, 0.00468]	0.1565	
vanilla	seq2seq	1000	-0.03208	[-0.0561, -0.00806]	0.0050	*
vanilla	seq2seq	1e+04	-0.04818	[-0.0722, -0.0242]	$< 10^{-4}$	*

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	5.014×10^{-4}	$[1.86 \times 10^{-4}, 8.30 \times 10^{-4}]$	0.6948	0.0267	0.8474
adaptive	seq2seq	-0.007179	[-0.00939, -0.00497]	0.0010	-0.432	0.0012
tropical	pointer	-4.229×10^{-4}	[-0.00405, 0.00262]	0.7966	-0.059	0.6763
tropical	seq2seq	2.379×10^{-4}	[-0.00202, 0.0022]	0.9175	0.00952	0.9512
vanilla	pointer	6.821×10^{-4}	$[3.66 \times 10^{-4}, 0.00108]$	0.1466	0.162	0.2272
vanilla	seq2seq	-0.00762	[-0.00975, -0.00597]	$< 10^{-4}$	-0.636	0.0002

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.03736	[0.0106, 0.0642]	0.2160
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.03752	[0.00625, 0.0688]	0.2386
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.03648	[0.00612, 0.0668]	0.2386
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.03838	[-0.00426, 0.081]	0.2673
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.03946	[0.00857, 0.0703]	0.2386
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.04614	[0.0244, 0.0679]	0.0665
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.03	[0.015, 0.045]	0.0722
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.02894	[0.00304, 0.0548]	0.2386
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.01368	[-0.0163, 0.0437]	0.8167

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.01762	[-0.0208, 0.0561]	0.8167	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.01314	[-0.0277, 0.054]	0.8167	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.03206	[0.00673, 0.0574]	0.2386	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.02872	[0.014, 0.0434]	0.0724	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.04006	[0.00998, 0.0701]	0.2296	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.05726	[0.00379, 0.111]	0.2386	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.05888	[0.0311, 0.0866]	0.0665	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00746	[-0.0147, 0.0296]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.0064	[-0.0098, 0.0226]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00636	[-0.0137, 0.0264]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.00496	[-0.0165, 0.0264]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.00322	[-0.0155, 0.022]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.00732	[-0.0138, 0.0285]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.00328	[-0.018, 0.0245]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.00598	[-0.00759, 0.0195]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.02424	[-0.0493, 8.69×10^{-4}]	0.8282	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.01498	[-0.0523, 0.0224]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.01554	[-0.0554, 0.0243]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.01428	[-0.0542, 0.0256]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.0138	[-0.0388, 0.0112]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.00466	[-0.0354, 0.0447]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.00246	[-0.0388, 0.0339]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.0259	[-0.0324, -0.0194]	0.0061	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.0346	[0.00912, 0.0601]	0.3923	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.02884	[-0.00208, 0.0598]	0.9103	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.03114	[-0.00311, 0.0654]	0.9105	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0287	[-0.00963, 0.067]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.02824	[-0.0089, 0.0654]	1.0000	

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.02788	[2.45×10^{-4}, 0.0555]	0.7802	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.04764	[0.0258, 0.0695]	0.0825	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.0898	[0.0672, 0.112]	0.0092	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.02658	[0.0018, 0.0514]	0.6941	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.02736	[-0.0295, 0.0842]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.03258	[-0.0179, 0.083]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01578	[-0.0351, 0.0667]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.02196	[-0.0235, 0.0674]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.0313	[0.00353, 0.0591]	0.6335	
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.01464	[-0.0298, 0.059]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.02798	[-0.0158, 0.0717]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0029	[-0.0128, 0.0186]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.00746	[-0.0112, 0.0261]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.00924	[-0.00997, 0.0285]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.00946	[-0.00915, 0.0281]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.01122	[-0.00217, 0.0246]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02522	[0.00397, 0.0465]	0.5717	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.0419	[0.0157, 0.0681]	0.2374	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.05792	[0.0408, 0.0751]	0.0166	*

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9487	[0.946, 0.951]	0.0028	5	0
adaptive	pointer	0.01	0.9495	[0.948, 0.951]	0.00188	5	0
adaptive	pointer	0.1	0.9495	[0.948, 0.951]	0.002	5	0
adaptive	pointer	1	0.9494	[0.948, 0.951]	0.0022	5	0
adaptive	pointer	10	0.9494	[0.947, 0.951]	0.00248	5	0
adaptive	pointer	100	0.9498	[0.948, 0.951]	0.00197	5	0
adaptive	pointer	1000	0.9493	[0.948, 0.951]	0.00183	5	0
adaptive	pointer	1e+04	0.9479	[0.946, 0.949]	0.00212	5	0
adaptive	seq2seq	0	0.9245	[0.922, 0.927]	0.00266	5	0
adaptive	seq2seq	0.01	0.9261	[0.924, 0.928]	0.00249	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	seq2seq	0.1	0.9261	[0.924, 0.928]	0.00252	5	0
adaptive	seq2seq	1	0.926	[0.924, 0.928]	0.00247	5	0
adaptive	seq2seq	10	0.9259	[0.924, 0.928]	0.00244	5	0
adaptive	seq2seq	100	0.926	[0.925, 0.927]	0.00133	5	0
adaptive	seq2seq	1000	0.9241	[0.923, 0.925]	0.00174	5	0
adaptive	seq2seq	1e+04	0.9151	[0.914, 0.916]	0.00148	5	0
tropical	pointer	0	0.9577	[0.957, 0.959]	0.00117	5	0
tropical	pointer	0.01	0.9587	[0.958, 0.959]	5.89×10^{-4}	5	0
tropical	pointer	0.1	0.959	[0.958, 0.96]	0.00107	5	0
tropical	pointer	1	0.9588	[0.957, 0.96]	0.00173	5	0
tropical	pointer	10	0.9591	[0.959, 0.959]	3.48×10^{-4}	5	0
tropical	pointer	100	0.9588	[0.958, 0.959]	6.53×10^{-4}	5	0
tropical	pointer	1000	0.9594	[0.959, 0.96]	0.00103	5	0
tropical	pointer	1e+04	0.9586	[0.958, 0.96]	0.00112	5	0
tropical	seq2seq	0	0.9464	[0.943, 0.95]	0.00418	5	0
tropical	seq2seq	0.01	0.9464	[0.943, 0.95]	0.00453	5	0
tropical	seq2seq	0.1	0.9464	[0.944, 0.95]	0.00388	5	0
tropical	seq2seq	1	0.9467	[0.943, 0.95]	0.00435	5	0
tropical	seq2seq	10	0.9468	[0.944, 0.95]	0.00415	5	0
tropical	seq2seq	100	0.9467	[0.944, 0.95]	0.00361	5	0
tropical	seq2seq	1000	0.9441	[0.942, 0.947]	0.00355	5	0
tropical	seq2seq	1e+04	0.9302	[0.926, 0.935]	0.00604	5	0
vanilla	pointer	0	0.9439	[0.942, 0.945]	0.00216	5	0
vanilla	pointer	0.01	0.9433	[0.941, 0.945]	0.00243	5	0
vanilla	pointer	0.1	0.9434	[0.941, 0.945]	0.00228	5	0
vanilla	pointer	1	0.9433	[0.941, 0.945]	0.00223	5	0
vanilla	pointer	10	0.9432	[0.941, 0.945]	0.00233	5	0
vanilla	pointer	100	0.9435	[0.942, 0.945]	0.00217	5	0
vanilla	pointer	1000	0.9437	[0.942, 0.945]	0.00219	5	0
vanilla	pointer	1e+04	0.9434	[0.942, 0.945]	0.00208	5	0
vanilla	seq2seq	0	0.9236	[0.923, 0.925]	0.0012	5	0
vanilla	seq2seq	0.01	0.9244	[0.923, 0.925]	0.00134	5	0
vanilla	seq2seq	0.1	0.9243	[0.923, 0.926]	0.0014	5	0
vanilla	seq2seq	1	0.9246	[0.924, 0.926]	0.00131	5	0
vanilla	seq2seq	10	0.9239	[0.923, 0.925]	0.00111	5	0
vanilla	seq2seq	100	0.9234	[0.923, 0.924]	5.50×10^{-4}	5	0
vanilla	seq2seq	1000	0.9223	[0.922, 0.922]	8.82×10^{-5}	5	0
vanilla	seq2seq	1e+04	0.9136	[0.912, 0.915]	0.00226	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	7.191×10^{-4}	[-0.00309, 0.00453]	0.9941
adaptive	pointer	0.1	7.586×10^{-4}	[-0.00305, 0.00457]	0.9920
adaptive	pointer	1	6.173×10^{-4}	[-0.00319, 0.00443]	0.9976
adaptive	pointer	10	6.920×10^{-4}	[-0.00312, 0.0045]	0.9953
adaptive	pointer	100	0.001012	[-0.0028, 0.00482]	0.9624
adaptive	pointer	1000	5.515×10^{-4}	[-0.00326, 0.00436]	0.9988
adaptive	pointer	1e+04	-8.832×10^{-4}	[-0.00469, 0.00293]	0.9814
adaptive	seq2seq	0.01	0.001601	[-0.00224, 0.00544]	0.7615
adaptive	seq2seq	0.1	0.001613	[-0.00223, 0.00545]	0.7558
adaptive	seq2seq	1	0.001458	[-0.00238, 0.0053]	0.8265
adaptive	seq2seq	10	0.001335	[-0.0025, 0.00517]	0.8756

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	seq2seq	100	0.001472	[-0.00237, 0.00531]	0.8203	
adaptive	seq2seq	1000	-4.503×10^{-4}	[-0.00429, 0.00339]	0.9997	
adaptive	seq2seq	1e+04	-0.009464	[-0.0133, -0.00563]	$< 10^{-4}$	*
tropical	pointer	0.01	0.001051	$[-7.69 \times 10^{-4}$, 0.00287]	0.4551	
tropical	pointer	0.1	0.001346	$[-4.74 \times 10^{-4}$, 0.00317]	0.2195	
tropical	pointer	1	0.001109	$[-7.11 \times 10^{-4}$, 0.00293]	0.4005	
tropical	pointer	10	0.001426	$[-3.94 \times 10^{-4}$, 0.00325]	0.1756	
tropical	pointer	100	0.001172	$[-6.49 \times 10^{-4}$, 0.00299]	0.3455	
tropical	pointer	1000	0.001784	$[-3.58 \times 10^{-5}$, 0.0036]	0.0565	
tropical	pointer	1e+04	9.638×10^{-4}	$[-8.56 \times 10^{-4}$, 0.00278]	0.5440	
tropical	seq2seq	0.01	-5.083×10^{-5}	[-0.00765, 0.00755]	1.0000	
tropical	seq2seq	0.1	-5.135×10^{-5}	[-0.00765, 0.00755]	1.0000	
tropical	seq2seq	1	2.997×10^{-4}	[-0.0073, 0.0079]	1.0000	
tropical	seq2seq	10	3.918×10^{-4}	[-0.00721, 0.00799]	1.0000	
tropical	seq2seq	100	2.377×10^{-4}	[-0.00736, 0.00783]	1.0000	
tropical	seq2seq	1000	-0.0023	[-0.0099, 0.0053]	0.9306	
tropical	seq2seq	1e+04	-0.01624	[-0.0238, -0.00865]	$< 10^{-4}$	*
vanilla	pointer	0.01	-5.813×10^{-4}	[-0.00449, 0.00332]	0.9986	
vanilla	pointer	0.1	-4.417×10^{-4}	[-0.00435, 0.00346]	0.9998	
vanilla	pointer	1	-5.504×10^{-4}	[-0.00446, 0.00336]	0.9990	
vanilla	pointer	10	-6.712×10^{-4}	[-0.00458, 0.00323]	0.9966	
vanilla	pointer	100	-3.850×10^{-4}	[-0.00429, 0.00352]	0.9999	
vanilla	pointer	1000	-1.823×10^{-4}	[-0.00409, 0.00372]	1.0000	
vanilla	pointer	1e+04	-4.465×10^{-4}	[-0.00435, 0.00346]	0.9997	
vanilla	seq2seq	0.01	7.106×10^{-4}	[-0.00157, 0.00299]	0.9208	
vanilla	seq2seq	0.1	6.191×10^{-4}	[-0.00166, 0.0029]	0.9581	
vanilla	seq2seq	1	9.092×10^{-4}	[-0.00137, 0.00319]	0.7937	
vanilla	seq2seq	10	2.644×10^{-4}	[-0.00201, 0.00254]	0.9997	
vanilla	seq2seq	100	-2.002×10^{-4}	[-0.00248, 0.00208]	1.0000	
vanilla	seq2seq	1000	-0.001327	[-0.00361, 9.52×10^{-4}]	0.4466	
vanilla	seq2seq	1e+04	-0.01002	[-0.0123, -0.00774]	$< 10^{-4}$	*

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-1.724×10^{-4}	$[-2.09 \times 10^{-4}, -1.34 \times 10^{-4}]$	0.3090	-0.128	0.3351
adaptive	seq2seq	-0.001332	[-0.00152, -0.00104]	$< 10^{-4}$	-0.429	0.0008
tropical	pointer	2.418×10^{-5}	$[-9.35 \times 10^{-5}, 1.65 \times 10^{-4}]$	0.7715	-0.00571	0.9768
tropical	seq2seq	-0.001898	[-0.0023, -0.00151]	0.0006	-0.356	0.0068
vanilla	pointer	3.887×10^{-5}	$[-1.07 \times 10^{-4}, 2.22 \times 10^{-4}]$	0.8258	0.0781	0.5567
vanilla	seq2seq	-0.001328	[-0.00145, -0.00121]	$< 10^{-4}$	-0.619	0.0002

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01377	[0.0106, 0.017]	0.0013	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.01541	[0.0126, 0.0182]	0.0013	*

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.01556	[0.0126, 0.0186]	0.0013	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.01543	[0.0113, 0.0195]	0.0013	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.01587	[0.0132, 0.0186]	0.0012	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01533	[0.0125, 0.0181]	0.0013	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.01574	[0.0121, 0.0194]	0.0013	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.01518	[0.0124, 0.018]	0.0013	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.02277	[0.0181, 0.0274]	0.0013	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.02201	[0.0176, 0.0265]	0.0013	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.0221	[0.0185, 0.0257]	0.0012	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.02217	[0.0175, 0.0268]	0.0013	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.0229	[0.0188, 0.027]	0.0013	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.02321	[0.0194, 0.0271]	0.0012	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.0218	[0.0174, 0.0262]	0.0013	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.01655	[0.011, 0.0221]	0.0013	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.00487	[0.00392, 0.00581]	0.0022	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.00617	[0.00365, 0.00869]	0.0221	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00607	[0.00396, 0.00818]	0.0133	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.006037	[0.00409, 0.00799]	0.0111	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.006233	[0.00434, 0.00813]	0.0095	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.006267	[0.00478, 0.00776]	0.0044	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.005603	[0.00429, 0.00692]	0.0044	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.004433	[0.00312, 0.00574]	0.0093	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	8.940×10^{-4}	[-0.00138, 0.00317]	0.3373	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.001784	$[-1.39 \times 10^{-4}, 0.00371]$	0.2891	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.001888	$[-1.01 \times 10^{-4}, 0.00388]$	0.2891	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.001443	$[-6.00 \times 10^{-4}, 0.00348]$	0.2891	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.001964	$[3.00 \times 10^{-5}, 0.0039]$	0.2871	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.002566	[0.00147, 0.00366]	0.0230	*

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.00177	$[-3.80 \times 10^{-4}, 0.00392]$	0.2891	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.001447	$[3.79 \times 10^{-4}, 0.00251]$	0.1381	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.02421	[0.0212, 0.0272]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.02333	[0.0214, 0.0253]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.02336	[0.0215, 0.0252]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.02337	[0.0218, 0.025]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.02357	[0.0218, 0.0254]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.02375	[0.0211, 0.0264]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.02522	[0.0224, 0.028]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.03279	[0.0309, 0.0347]	$< 10^{-4}$	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.01124	[0.00658, 0.0159]	0.0155	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.01234	[0.00643, 0.0182]	0.0155	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.01263	[0.00723, 0.018]	0.0155	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01205	[0.00564, 0.0184]	0.0155	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.01227	[0.00687, 0.0177]	0.0155	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.01217	[0.00696, 0.0174]	0.0155	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.01532	[0.01, 0.0206]	0.0091	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.02844	[0.0204, 0.0365]	0.0048	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.02024	[0.0179, 0.0225]	0.0002	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.01895	[0.0167, 0.0212]	0.0002	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.01918	[0.0174, 0.021]	0.0001	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.01878	[0.0167, 0.0208]	0.0002	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.0193	[0.0173, 0.0213]	0.0002	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.02005	[0.0177, 0.0225]	0.0002	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.02138	[0.0187, 0.0241]	0.0002	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.02981	[0.0286, 0.031]	$< 10^{-4}$	*

A.7 id_exact

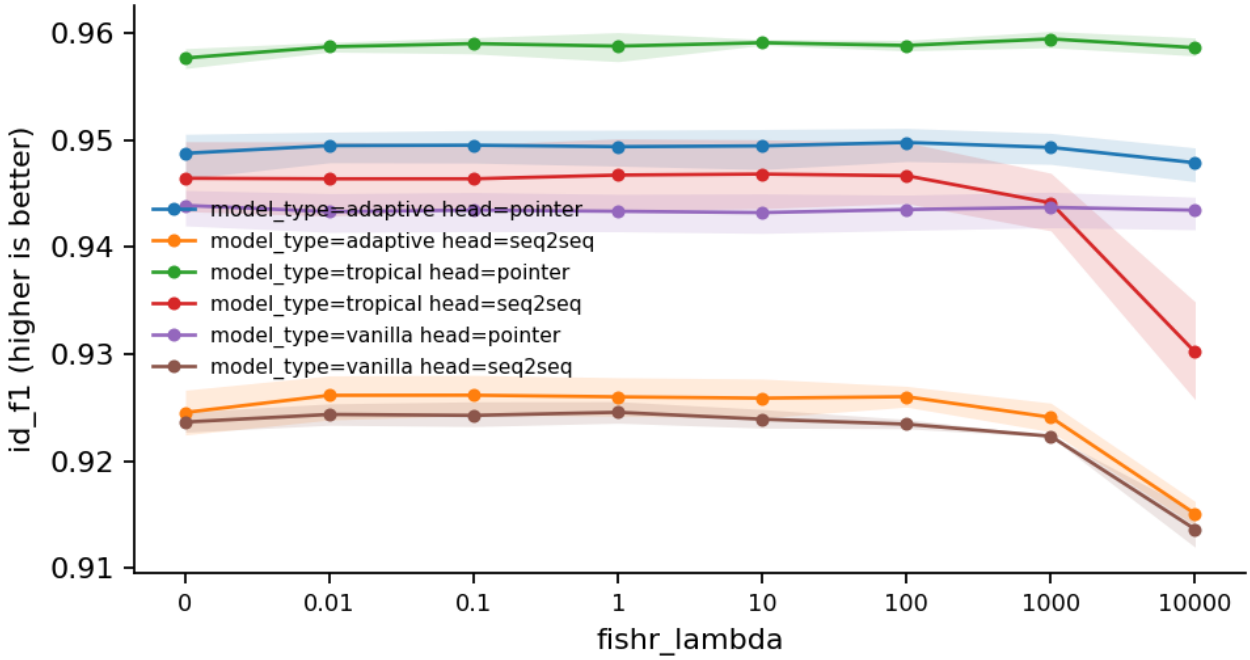


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

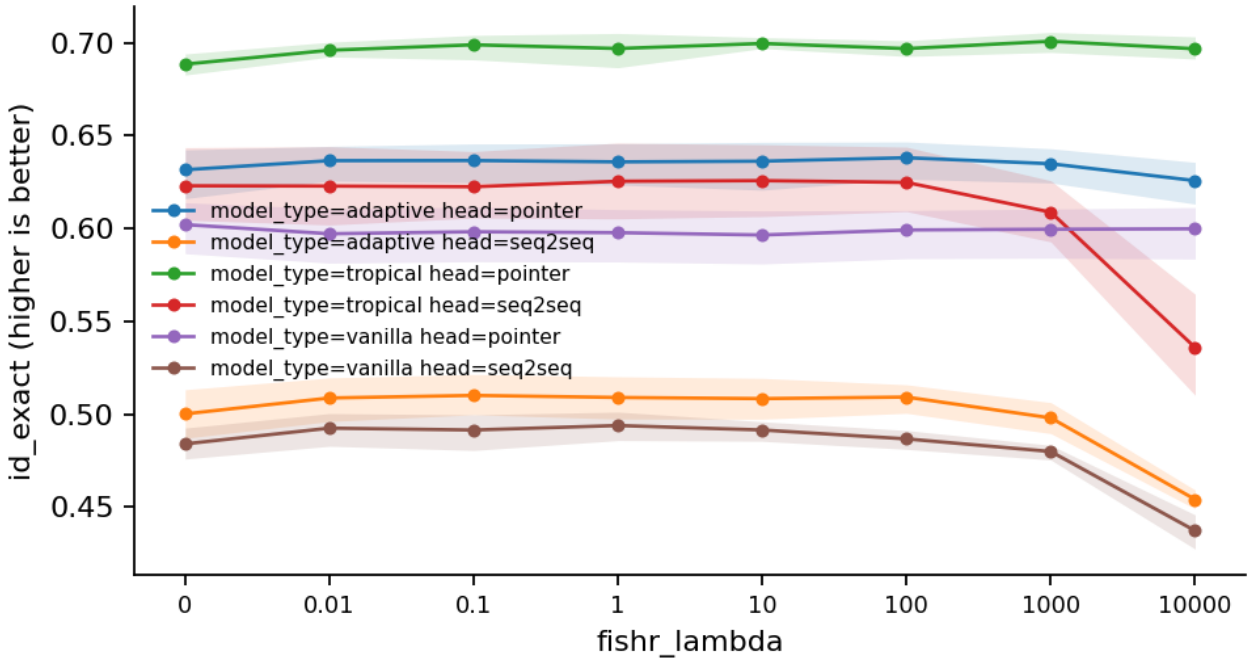


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6317	[0.616, 0.642]	0.018	5	0
adaptive	pointer	0.01	0.6365	[0.626, 0.644]	0.0125	5	0
adaptive	pointer	0.1	0.6366	[0.625, 0.646]	0.0136	5	0
adaptive	pointer	1	0.6359	[0.623, 0.646]	0.0144	5	0
adaptive	pointer	10	0.6363	[0.621, 0.646]	0.0172	5	0
adaptive	pointer	100	0.6381	[0.627, 0.647]	0.0127	5	0
adaptive	pointer	1000	0.6349	[0.625, 0.643]	0.0115	5	0
adaptive	pointer	1e+04	0.6258	[0.613, 0.636]	0.0146	5	0
adaptive	seq2seq	0	0.5002	[0.487, 0.513]	0.0167	5	0
adaptive	seq2seq	0.01	0.5088	[0.496, 0.52]	0.0145	5	0
adaptive	seq2seq	0.1	0.5102	[0.5, 0.522]	0.0148	5	0
adaptive	seq2seq	1	0.5091	[0.498, 0.52]	0.0154	5	0
adaptive	seq2seq	10	0.5084	[0.497, 0.519]	0.0146	5	0
adaptive	seq2seq	100	0.5093	[0.501, 0.516]	0.01	5	0
adaptive	seq2seq	1000	0.4981	[0.49, 0.506]	0.0108	5	0
adaptive	seq2seq	1e+04	0.4542	[0.449, 0.459]	0.00652	5	0
tropical	pointer	0	0.6885	[0.683, 0.694]	0.00741	5	0
tropical	pointer	0.01	0.696	[0.692, 0.7]	0.00502	5	0
tropical	pointer	0.1	0.6989	[0.691, 0.704]	0.00894	5	0
tropical	pointer	1	0.6969	[0.687, 0.705]	0.0119	5	0
tropical	pointer	10	0.6996	[0.697, 0.703]	0.00374	5	0
tropical	pointer	100	0.6969	[0.693, 0.701]	0.0057	5	0
tropical	pointer	1000	0.7008	[0.695, 0.706]	0.00694	5	0
tropical	pointer	1e+04	0.6968	[0.691, 0.703]	0.00744	5	0
tropical	seq2seq	0	0.623	[0.605, 0.644]	0.0248	5	0
tropical	seq2seq	0.01	0.6229	[0.602, 0.644]	0.0274	5	0
tropical	seq2seq	0.1	0.6225	[0.606, 0.642]	0.0236	5	0
tropical	seq2seq	1	0.6255	[0.605, 0.646]	0.0268	5	0
tropical	seq2seq	10	0.6258	[0.607, 0.645]	0.025	5	0
tropical	seq2seq	100	0.6248	[0.609, 0.644]	0.0219	5	0
tropical	seq2seq	1000	0.6088	[0.593, 0.626]	0.0217	5	0
tropical	seq2seq	1e+04	0.536	[0.51, 0.565]	0.0349	5	0
vanilla	pointer	0	0.6021	[0.587, 0.614]	0.0177	5	0
vanilla	pointer	0.01	0.5972	[0.581, 0.611]	0.0192	5	0
vanilla	pointer	0.1	0.5983	[0.582, 0.612]	0.0182	5	0
vanilla	pointer	1	0.5978	[0.582, 0.61]	0.0179	5	0
vanilla	pointer	10	0.5965	[0.581, 0.61]	0.0184	5	0
vanilla	pointer	100	0.5992	[0.584, 0.61]	0.0173	5	0
vanilla	pointer	1000	0.5996	[0.584, 0.611]	0.0179	5	0
vanilla	pointer	1e+04	0.5999	[0.584, 0.611]	0.019	5	0
vanilla	seq2seq	0	0.4841	[0.476, 0.493]	0.0108	5	0
vanilla	seq2seq	0.01	0.4926	[0.483, 0.5]	0.0112	5	0
vanilla	seq2seq	0.1	0.4915	[0.481, 0.5]	0.0123	5	0
vanilla	seq2seq	1	0.494	[0.486, 0.501]	0.0107	5	0
vanilla	seq2seq	10	0.4916	[0.486, 0.496]	0.00685	5	0
vanilla	seq2seq	100	0.4868	[0.481, 0.491]	0.0068	5	0
vanilla	seq2seq	1000	0.48	[0.475, 0.483]	0.00519	5	0
vanilla	seq2seq	1e+04	0.4373	[0.428, 0.446]	0.0117	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.00488	[-0.0204, 0.0301]	0.9932	
adaptive	pointer	0.1	0.00496	[-0.0203, 0.0302]	0.9926	
adaptive	pointer	1	0.00424	[-0.021, 0.0295]	0.9970	
adaptive	pointer	10	0.0046	[-0.0207, 0.0299]	0.9952	
adaptive	pointer	100	0.00648	[-0.0188, 0.0317]	0.9684	
adaptive	pointer	1000	0.00324	[-0.022, 0.0285]	0.9994	
adaptive	pointer	1e+04	-0.00586	[-0.0311, 0.0194]	0.9813	
adaptive	seq2seq	0.01	0.00856	[-0.0147, 0.0318]	0.8455	
adaptive	seq2seq	0.1	0.00996	[-0.0133, 0.0332]	0.7410	
adaptive	seq2seq	1	0.00884	[-0.0144, 0.0321]	0.8262	
adaptive	seq2seq	10	0.00822	[-0.015, 0.0315]	0.8676	
adaptive	seq2seq	100	0.00906	[-0.0142, 0.0323]	0.8103	
adaptive	seq2seq	1000	-0.00216	[-0.0254, 0.0211]	0.9999	
adaptive	seq2seq	1e+04	-0.04602	[-0.0693, -0.0228]	$< 10^{-4}$	*
tropical	pointer	0.01	0.0075	[-0.00563, 0.0206]	0.4660	
tropical	pointer	0.1	0.01044	[-0.00269, 0.0236]	0.1647	
tropical	pointer	1	0.00842	[-0.00471, 0.0215]	0.3488	
tropical	pointer	10	0.01114	[-0.00199, 0.0243]	0.1232	
tropical	pointer	100	0.00838	[-0.00475, 0.0215]	0.3535	
tropical	pointer	1000	0.01232	$[-8.06 \times 10^{-4}, 0.0254]$	0.0730	
tropical	pointer	1e+04	0.00828	[-0.00485, 0.0214]	0.3653	
tropical	seq2seq	0.01	-1.400×10^{-4}	[-0.0457, 0.0454]	1.0000	
tropical	seq2seq	0.1	-5.000×10^{-4}	[-0.0461, 0.0451]	1.0000	
tropical	seq2seq	1	0.00244	[-0.0431, 0.048]	1.0000	
tropical	seq2seq	10	0.00278	[-0.0428, 0.0483]	1.0000	
tropical	seq2seq	100	0.00182	[-0.0437, 0.0474]	1.0000	
tropical	seq2seq	1000	-0.0142	[-0.0598, 0.0314]	0.9210	
tropical	seq2seq	1e+04	-0.08704	[-0.133, -0.0415]	$< 10^{-4}$	*
vanilla	pointer	0.01	-0.00492	[-0.0367, 0.0269]	0.9982	
vanilla	pointer	0.1	-0.00384	[-0.0356, 0.0279]	0.9996	
vanilla	pointer	1	-0.00434	[-0.0361, 0.0274]	0.9992	
vanilla	pointer	10	-0.00558	[-0.0374, 0.0262]	0.9961	
vanilla	pointer	100	-0.00288	[-0.0347, 0.0289]	0.9999	
vanilla	pointer	1000	-0.0025	[-0.0343, 0.0293]	1.0000	
vanilla	pointer	1e+04	-0.00226	[-0.034, 0.0295]	1.0000	
vanilla	seq2seq	0.01	0.00848	[-0.00861, 0.0256]	0.6088	
vanilla	seq2seq	0.1	0.00746	[-0.00963, 0.0246]	0.7251	
vanilla	seq2seq	1	0.00988	[-0.00721, 0.027]	0.4540	
vanilla	seq2seq	10	0.00748	[-0.00961, 0.0246]	0.7227	
vanilla	seq2seq	100	0.00268	[-0.0144, 0.0198]	0.9980	
vanilla	seq2seq	1000	-0.00412	[-0.0212, 0.013]	0.9771	
vanilla	seq2seq	1e+04	-0.04674	[-0.0638, -0.0296]	$< 10^{-4}$	*

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.001194	[-0.0014, -0.00105]	0.2904	-0.149	0.2559
adaptive	seq2seq	-0.006706	[-0.00785, -0.00497]	0.0001	-0.383	0.0036
tropical	pointer	2.164×10^{-4}	$[-5.19 \times 10^{-4}, 0.00104]$	0.7218	-0.00381	0.9872
tropical	seq2seq	-0.01031	[-0.0126, -0.00797]	0.0009	-0.354	0.0066
vanilla	pointer	4.329×10^{-4}	$[-8.17 \times 10^{-4}, 0.002]$	0.7630	0.109	0.4117

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
vanilla	seq2seq	-0.007001	[-0.0076, -0.00596]	$< 10^{-4}$	-0.566	0.0002

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.08638	[0.0612, 0.112]	0.0029	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.01	0.0988	[0.0766, 0.121]	0.0029	*
model_type: tropical - vanilla head=pointer fishr_lambda=0.1	0.1007	[0.0755, 0.126]	0.0029	*
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.09914	[0.0681, 0.13]	0.0029	*
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1031	[0.0811, 0.125]	0.0028	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.09764	[0.0759, 0.119]	0.0029	*
model_type: tropical - vanilla head=pointer fishr_lambda=1000	0.1012	[0.073, 0.129]	0.0029	*
model_type: tropical - vanilla head=pointer fishr_lambda=1e+04	0.09692	[0.0715, 0.122]	0.0029	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.1389	[0.105, 0.172]	0.0029	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.1303	[0.1, 0.161]	0.0029	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.131	[0.105, 0.157]	0.0024	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.1315	[0.1, 0.163]	0.0029	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.1342	[0.105, 0.164]	0.0029	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.1381	[0.114, 0.162]	0.0014	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.1289	[0.101, 0.157]	0.0029	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.09864	[0.0646, 0.133]	0.0029	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.02954	[0.0258, 0.0332]	0.0004	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.03934	[0.0222, 0.0564]	0.0246	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.03834	[0.0234, 0.0533]	0.0192	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.03812	[0.0252, 0.051]	0.0133	*
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.03972	[0.0266, 0.0528]	0.0130	*
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.0389	[0.0279, 0.0499]	0.0091	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.03528	[0.0243, 0.0462]	0.0112	*
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.02594	[0.016, 0.0359]	0.0192	*

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.01614	[-0.00418, 0.0365]	0.0921	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.01622	[0.00839, 0.024]	0.0317	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.01864	[0.00916, 0.0281]	0.0329	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.0151	[0.00456, 0.0256]	0.0647	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01688	[0.00471, 0.0291]	0.0647	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.02252	[0.0161, 0.0289]	0.0091	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.0181	[0.00553, 0.0307]	0.0647	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.01686	[0.00792, 0.0258]	0.0329	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.1314	[0.111, 0.152]	0.0009	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.1278	[0.115, 0.14]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.1264	[0.115, 0.138]	0.0001	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1268	[0.116, 0.138]	0.0001	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.1278	[0.116, 0.14]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.1289	[0.109, 0.149]	0.0008	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.1368	[0.122, 0.151]	0.0002	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.1716	[0.156, 0.187]	0.0001	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.06548	[0.0367, 0.0943]	0.0193	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.01	0.07312	[0.0348, 0.111]	0.0193	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0.1	0.07642	[0.04, 0.113]	0.0193	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.07146	[0.0294, 0.113]	0.0193	*
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.07384	[0.0395, 0.108]	0.0193	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.07204	[0.0388, 0.105]	0.0193	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1000	0.092	[0.058, 0.126]	0.0117	*
head: pointer - seq2seq model_type=tropical fishr_lambda=1e+04	0.1608	[0.114, 0.208]	0.0055	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.118	[0.0973, 0.139]	0.0010	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1046	[0.0881, 0.121]	0.0009	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1067	[0.0933, 0.12]	0.0004	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1038	[0.0895, 0.118]	0.0006	*

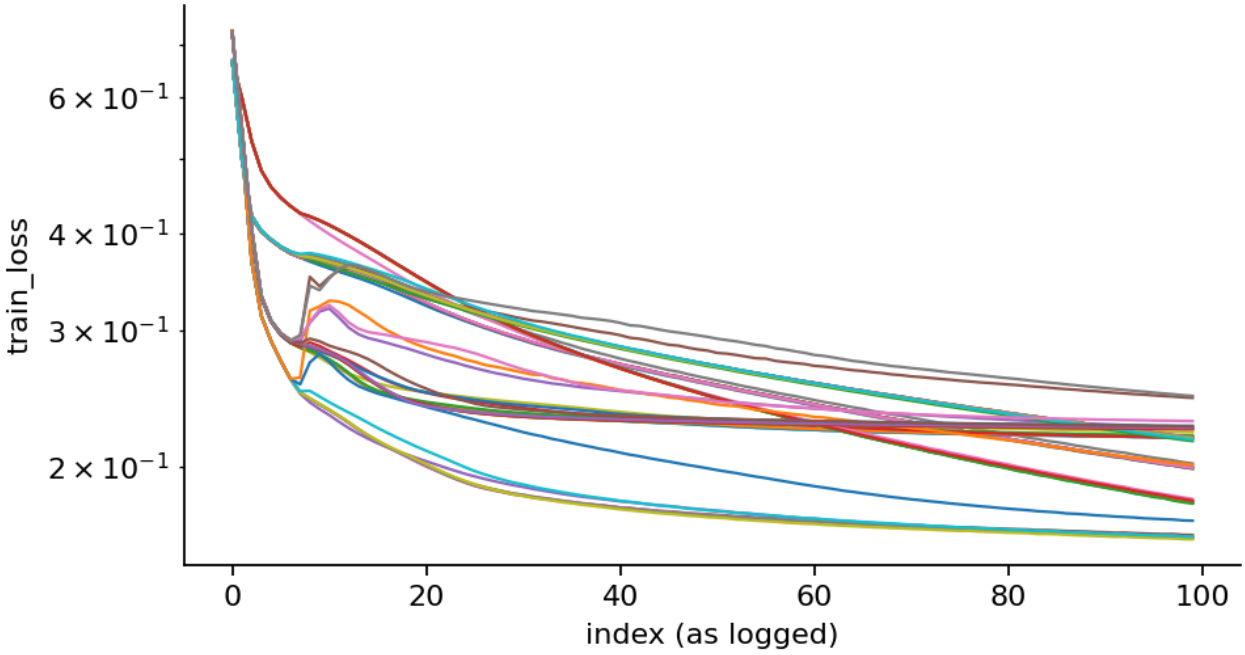


Figure 9: `train_loss`, the mean over seeds per cell.

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.105	[0.0875, 0.122]	0.0009	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1125	[0.0904, 0.135]	0.0015	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.1197	[0.0946, 0.145]	0.0017	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.1625	[0.152, 0.173]	$< 10^{-4}$	*

B Curves

C Configuration

The base configuration every run shares (frozen hash `3b4ea5b21c9e`); factor levels are in Table 1.

```
task:
  name: FractionalKnapsackDataset
  value_range:
  - 1
  - 10
  target_range:
  - 10
  - 20
  weight_range:
  - 1
  - 10
  adversarial_range:
  - 10
```

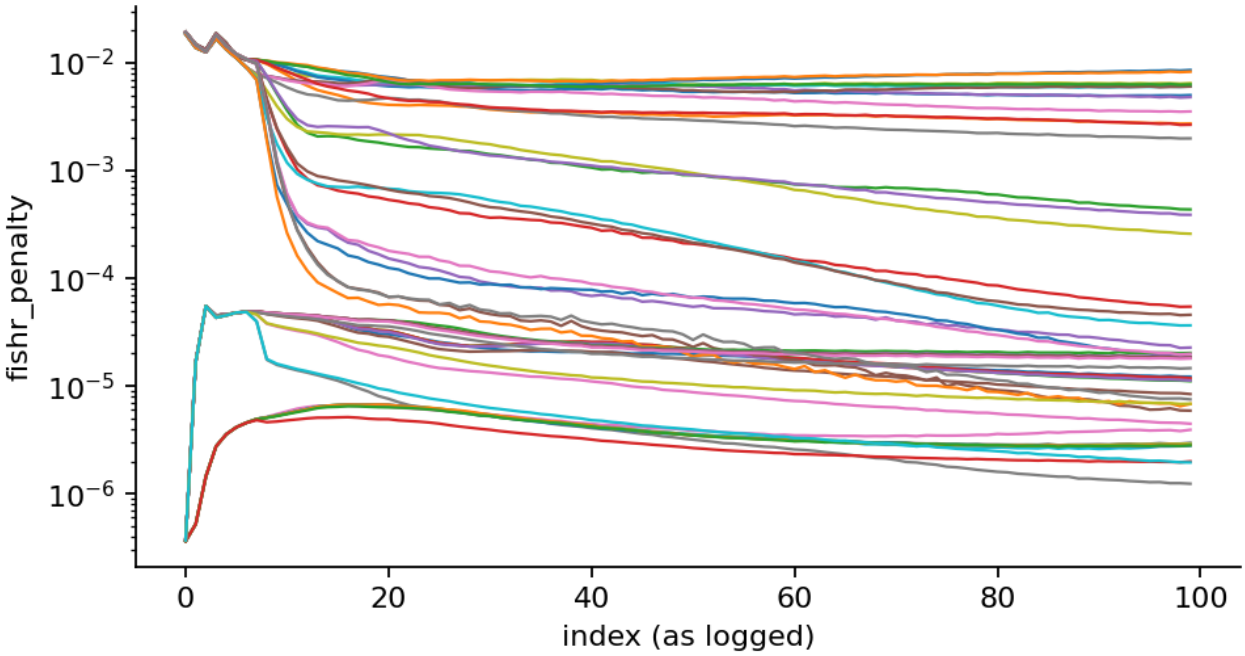


Figure 10: `fishr_penalty`, the mean over seeds per cell.

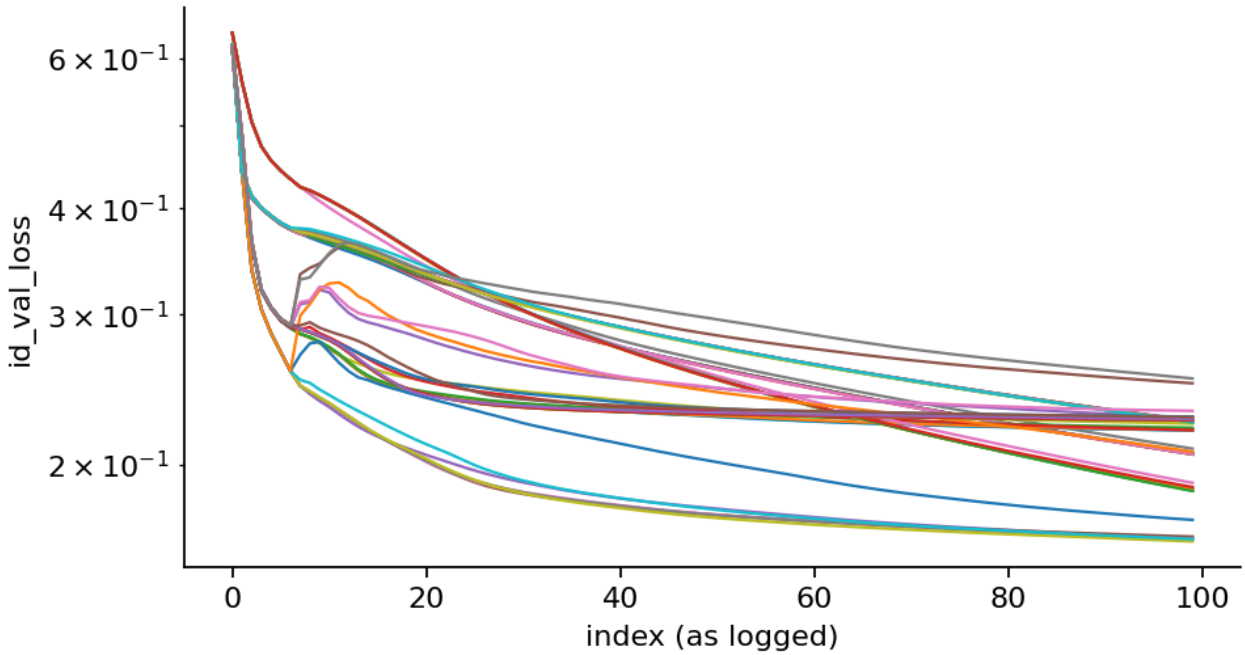


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 21
  ood_noise_range:
  - 1
  - 5
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 100
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/FractionalKnapsack/results`.